

* Look at BPS solutions to classical gravity
BPS states: Designate a solution of the theory that is invariant under a subalg. of the SUSY alg. of the action that contains at least one fermionic generator.

→ Can be obtained by solving first order ODE w/ the ferm. transf. rules of the theory.

→ For each preserved susy, $\exists$ a Killing spinor and a subset of Killing Vectors of the geometry

1. Frames/Connections on the Sphere:

Differential Geometry of S^n :

(i) There are frame 1-forms $\bar{e}^a$, $a=1, \dots, n$ s.t. the line elements of S^n can be written as

$$d\Omega_n^2 = \delta_{ab} \bar{e}^a \bar{e}^b$$

$$\bar{e}^a = e^a_m dx^m$$

(ii) There are torsion-free connection 1-forms $\bar{\omega}^{ab}$, which satisfy the Cartan structure equation

$$d\bar{e}^a + \bar{\omega}^a_b \wedge \bar{e}^b = T^a = 0$$

(iii) S^n is maximally symmetric, so the curvature 2-form

$$\rho^{ab} = d\bar{\omega}^{ab} + \bar{\omega}^a_c \wedge \bar{\omega}^{cb}$$

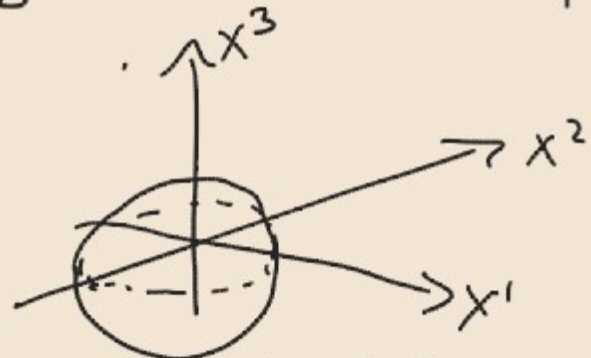
$$\rho^{ab} = \frac{1}{2} R_{mn}{}^{ab}(x) dx^m \wedge dx^n$$

→ Maximal symmetry: $R_{abcd} = \frac{R}{n(n-1)} [g_{ac}g_{bd} - g_{ad}g_{bc}]$

$$\begin{aligned}
 \therefore g^{ab} &= \frac{1}{2} e^a_{\alpha} e^b_{\beta} R^{\alpha\beta}{}_{\mu\nu} dx^{\mu} dx^{\nu} \\
 &= \frac{1}{2} e^a_{\alpha} e^b_{\beta} \left(\frac{R}{n(n-1)} \right) [g_{\mu\alpha} g_{\nu\beta} - g_{\mu\beta} g_{\nu\alpha}] dx^{\mu} dx^{\nu} \\
 &= e^a_{\alpha} e^b_{\beta} \frac{R}{n(n-1)} g_{\mu\alpha} g_{\nu\beta} dx^{\mu} dx^{\nu} \\
 &= e^a_{\alpha} e^b_{\beta} \left(\frac{R}{n(n-1)} \right) e^{\alpha}_{\sigma} \delta^{\sigma}_{\mu} e^{\beta}_{\tau} \delta^{\tau}_{\nu} \dots \\
 &= e^a_{\alpha} e^b_{\beta} g^{\alpha\sigma} g^{\beta\tau} \left(\frac{R}{n(n-1)} \right) e^{\sigma}_{\mu} \delta^{\mu}_{\sigma} e^{\tau}_{\nu} \delta^{\nu}_{\tau} e^{\epsilon}_{\rho} \delta^{\rho}_{\epsilon} e^f_{\lambda} \delta^{\lambda}_{\epsilon} dx^{\mu} dx^{\nu} \\
 &= \delta^a_{\sigma} \delta^b_{\tau} \frac{R}{n(n-1)} \delta^{\sigma}_{\mu} e^{\mu}_{\epsilon} e^{\epsilon}_{\nu} \delta^{\tau}_{\nu} dx^{\mu} dx^{\nu} \\
 &= \delta^a_{\mu} \delta^b_{\nu} \left(\frac{R}{n(n-1)} \right) e^{\mu}_{\epsilon} e^{\nu}_{\tau} dx^{\mu} dx^{\nu} \\
 &= \left[\frac{R}{n(n-1)} e^a_{\mu} e^b_{\nu} dx^{\mu} dx^{\nu} \right] \\
 \bar{e}^a &\equiv \frac{R}{n(n-1)} e^a_{\mu} dx^{\mu}
 \end{aligned}$$

*Idea: Use Hyperspherical coordinates to generate the frame fields on S^n .

$$S^2: (x^1)^2 + (x^2)^2 + (x^3)^2 = 1$$



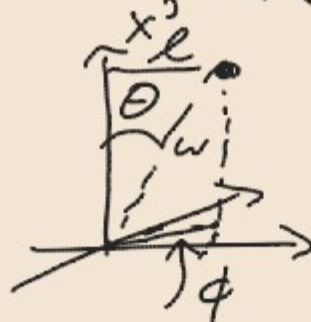
$$x^3 = \cos(\theta) r$$

$$x^2 = \sin(\phi) l$$

$$x^2 = \sin(\phi) \sin(\theta) r$$

$$x^1 = \cos(\phi) \sin(\theta) r$$

$$\hookrightarrow \mathbb{R}^3$$



$$r = 1$$

$$\sin(\theta) = l$$

$$\therefore x^2 = \cos(\phi) \sin(\theta) \quad , \quad x^1 = \sin(\phi) \sin(\theta)$$

$$x^3 = \cos(\theta^1)$$

$$x^2 = "x", \quad x^1 = "y"$$

$$x^3 = z$$

$$\phi = \theta^1, \quad \theta = \theta^2$$

$$0 \leq \theta^1 \leq 2\pi, \quad 0 \leq \theta^2 \leq \pi$$

$$\therefore d\Omega_{(n)}^2 = (dx^1)^2 + (dx^2)^2 + (dx^3)^2$$

$$dx^2 = d(\cos(\theta^1) \sin(\theta^2)) = -\sin(\theta^1) d\theta^1 \sin(\theta^2) + \cos(\theta^1) \cos(\theta^2) d\theta^2$$

$$(dx^2)^2 = \sin^2(\theta^1) (d\theta^1)^2 + \cos^2(\theta^1) (d\theta^2)^2$$

etc...

$$d\Omega_{(n)}^2 = (d\theta^1)^2 + \sin^2(\theta^2) (d\theta^2)^2$$

Define: $\bar{e}^2 := d\theta^2$

$$\bar{e}^1 = \sin(\theta^2) d\theta^1$$

Then, from

$$d\bar{e}^a + \bar{\omega}^a_b \wedge \bar{e}^b = 0$$

$$d(\bar{e}^2) = 0, \quad \bar{\omega}^2_1 \wedge \bar{e}^1 + \bar{\omega}^2_2 \wedge \bar{e}^2$$

$$\bar{\omega}^{12} = \cos(\theta^2) d\theta^1$$

Recursive Relation: $S^n: \sum_{i=1}^{n+1} (x_{(n)}^i)^2 = 1$
embedded in $\mathbb{R}^{n+1}$.

• $x_{(n)}^i: \theta^n$

$x_{(n-1)}^i$ of S^{n-1} by:

$$\theta^0 \equiv 1$$

$$x_{(n)}^{n+1} = \cos(\theta^n)$$

$$0 \leq \theta^n \leq \pi$$

$$x_{(n)}^a = \sin(\theta^n) x_{(n-1)}^a$$

$$a \leq n$$

• Explicitly defines angular coordinates $\theta^a, a=1, \dots, n$

$$0 \leq \theta^1 \leq 2\pi, \quad 0 \leq \theta^a \leq \pi$$

• S^3 : $X_{(3)}^4 = \cos(\theta^3)$

$$\begin{cases} X_{(3)}^3 = \sin(\theta^3) X_{(2)}^3 \\ \hookrightarrow X_{(2)}^3 = \cos(\theta^2) \\ X_{(2)}^2 = \sin(\theta^3) \cos(\theta^2) \\ X_{(3)}^2 = \sin(\theta^3) X_{(2)}^2 \\ \hookrightarrow X_{(2)}^2 = \sin(\theta^2) X_{(1)}^2 = \sin(\theta^2) \cos(\theta^1) \\ X_{(3)}^1 = \sin(\theta^3) \sin(\theta^2) \cos(\theta^1) \\ X_{(3)}^2 = \sin(\theta^3) X_{(2)}^2 = \sin(\theta^3) \sin(\theta^2) X_{(1)}^2 \\ = \sin(\theta^3) \sin(\theta^2) \sin(\theta^1) X_{(1)}^1 \end{cases}$$

Line element of S^n :

$$d\Omega_{(n)}^2 = (d\theta^n)^2 + \sin^2(\theta^n) d\Omega_{(n-1)}^2$$

Frame & connection frames:

$$\begin{aligned} \bar{e}_{(n)}^n &= d\theta^n, & \bar{e}_{(n)}^a &= \sin(\theta^n) \bar{e}_{(n-1)}^a, \quad a \leq n-1 \\ \bar{\omega}_{(n)}^{ab} &= \bar{\omega}_{(n-1)}^{ab}, & \bar{\omega}_{(n)}^{an} &= \cos(\theta^n) \bar{e}_{(n-1)}^a \end{aligned}$$

More explicitly:

$$\begin{aligned} \bar{e}^a &= \left(\prod_{j=a+1}^n \sin(\theta^j) \right) d\theta^a, \quad a \leq n \\ \bar{\omega}^{ab} &= \cos(\theta^b) \left(\prod_{j=a+1}^{b-1} \sin(\theta^j) \right) d\theta^a, \quad 1 \leq a < b \leq n \end{aligned}$$

Expansion: $\bar{e}_{(n)}^n = d\theta^n$

$$\bar{e}_{(n-1)}^{n-1} = \sin(\theta^n) \bar{e}_{(n-2)}^{n-1} = \sin(\theta^n) d\theta^{n-1}$$

$$e_{(a)}^{n-2} = \sinh(\theta^n) e_{(n-2)}^{n-2} = \sinh(\theta^n) \sinh(\theta^{n-1}) e_{(n-2)}^{n-2} \\ = \sinh(\theta^n) \sinh(\theta^{n-2}) d\theta^{n-2}$$

Claim: The curvature form $\bar{\omega}^{ab} = \bar{e}^a \wedge \bar{e}^b$ is obtained from ω^{ab} :

Proof:

$$\bar{\omega}^{ab} = d\bar{\omega}^{ab} + \omega_a^c \wedge \omega_c^b \\ d\bar{\omega}^{ab} = d \left\{ \cos(\theta^b) \left(\prod_{j=a+1}^{b-2} \sinh(\theta^j) \right) d\theta^a \right\} \\ = -\sinh(\theta^b) \left(\prod_{j=a+1}^{b-2} \sinh(\theta^j) \right) d\theta^b d\theta^a + \cos(\theta^b) \left(\prod_{j=a+1}^{b-2} \cosh(\theta^j) \right) d\theta^j d\theta^b \\ = -\sinh(\theta^b) \left(\prod_{j=a+1}^{b-2} \sinh(\theta^j) \right) d\theta^b d\theta^a \\ + \sum_{j=a+1}^{b-2} \cos(\theta^b) \left(\prod_{i=a+1}^{j-1} \sinh(\theta^i) \right) d\theta^i d\theta^b d\theta^c$$

$$\textcircled{b} \text{ Need } \omega_a^c \wedge \omega_c^b \\ \text{Need } \left(\cos(\theta^c) \left(\prod_{j=a+1}^{c-1} \sinh(\theta^j) \right) d\theta^a \right) \wedge \left(\cos(\theta^b) \left(\prod_{i=c+1}^{b-1} \sinh(\theta^i) \right) d\theta^c \right) \\ = \cos(\theta^c) \cos(\theta^b) \left(\prod_{j=a+1}^{c-1} \sinh(\theta^j) \right) \left(\prod_{i=c+1}^{b-1} \sinh(\theta^i) \right) d\theta^a d\theta^c$$

... not sure

should be

$$\bar{\omega}^{ab} = \left(\prod_{j=a+1}^n \sinh(\theta^j) \right) d\theta^a \wedge \left(\prod_{i=b+1}^n \sinh(\theta^i) \right) d\theta^b$$

Claim: The Volume of S^n is $\frac{2\pi^{(n+1)/2}}{\Gamma((n+1)/2)}$

$$\Omega_n \equiv \int_{S^n} \prod_{a=1}^n d\theta^a \sqrt{g} = \int \bar{e}^1 \wedge \dots \wedge \bar{e}^n = \frac{2\pi^{(n+1)/2}}{\Gamma((n+1)/2)}$$

Proof: Recall, that the volume element

$$\begin{aligned}
 dV &\equiv e^0 \wedge e^1 \wedge \dots \wedge e^{D-1} \\
 &= \frac{1}{D!} \epsilon_{a_1 \dots a_D} e^{a_1} \wedge \dots \wedge e^{a_D} \\
 &= \frac{1}{D!} e \epsilon_{\mu_1 \dots \mu_D} dx^{\mu_1} \wedge \dots \wedge dx^{\mu_D} \\
 &= e dx^0 \dots dx^{D-1} \\
 &= \sqrt{-g} d^D x \\
 \text{and } g &= e \bar{e} e \Rightarrow e = \sqrt{g}
 \end{aligned}$$

$$\begin{aligned}
 &\int \bar{e}^1 \wedge \dots \wedge \bar{e}^n \\
 &= \int \left(\left(\prod_{j=2}^n \sin(\theta_j) \right) d\theta^1 \wedge \left(\prod_{j=3}^n d\theta^2 \wedge \dots \wedge (\sin(\theta^n) d\theta^{n-1}) \right) \right) \\
 &= \int d\theta^1 \wedge (\sin(\theta^2) d\theta^2) \wedge (\sin(\theta^3)^2 d\theta^3) \wedge \dots \\
 &\quad \wedge (\sin(\theta^n)^{n-2} d\theta^n) \\
 &= (2\pi) \left(\int \sin(\theta^2) d\theta^2 \right) \left(\int \sin(\theta^3)^2 d\theta^3 \right) \dots
 \end{aligned}$$

Now, we use the Beta $\int^a$:

$$\begin{aligned}
 B(x, y) &\equiv \int_0^1 t^{x-1} (1-t)^{y-1} dt \\
 &= 2 \int_0^{\pi/2} \sin(\theta)^{2x-1} \cos(\theta)^{2y-1} d\theta
 \end{aligned}$$

$$\theta' = 2\theta$$

$$B(x, y) = \int_0^{\pi} \sin\left(\frac{\theta}{2}\right)^{2x-1} \cos\left(\frac{\theta}{2}\right)^{2y-1} d\theta$$

$$B(x, \frac{1}{2}) = \int_0^{\pi} \sin\left(\frac{\theta}{2}\right)^{2x-1} d\theta$$

$$x = (L+1)/2$$

$$B(L, \frac{1}{2}) = \int_0^{\pi} \sin\left(\frac{\theta}{2}\right)^L d\theta = \frac{\Gamma(L) \Gamma(\frac{1}{2})}{\Gamma(L+\frac{1}{2})}$$

$$2\pi \frac{\Gamma(\frac{1}{2})^{n-1} \Gamma(1) \Gamma(2) \dots \Gamma(n-2)}{\Gamma(\frac{3}{2}) \dots \Gamma(n-\frac{1}{2})}$$

$$2\pi \frac{\Gamma(\frac{1}{2})^{n-1}}{\Gamma(\frac{n+1}{2}) \Gamma(\frac{1}{2})}$$

$$B(\frac{n+1}{2}, \frac{1}{2}) = \int_0^{\pi} \sin^n(\theta) d\theta = \frac{\Gamma(\frac{n+1}{2}) \Gamma(\frac{1}{2})}{\Gamma(\frac{n+2}{2})}$$

$$2\pi \frac{\Gamma(\frac{1}{2})^{n-1}}{\Gamma(1) \Gamma(2) \dots \Gamma(n-2) \Gamma(n-1)}$$

$$\begin{aligned} & \left(\frac{\Gamma(\frac{3}{2})}{\Gamma(\frac{1}{2})} \right) \left(\frac{\Gamma(\frac{5}{2})}{\Gamma(\frac{3}{2})} \right) \left(\frac{\Gamma(\frac{7}{2})}{\Gamma(\frac{5}{2})} \right) \left(\frac{\Gamma(\frac{9}{2})}{\Gamma(\frac{7}{2})} \right) \dots \times \left(\frac{\Gamma(\frac{n}{2})}{\Gamma(\frac{n-1}{2})} \right) \\ & \Omega_n = \frac{2\pi (\sqrt{\pi})^n}{\Gamma(\frac{n+1}{2})} = \frac{2\pi^{\frac{n+1}{2}}}{\Gamma(\frac{n+1}{2})} \\ & \boxed{\Omega_n = \text{Vol}(S^n) = \frac{2\pi^{\frac{n+1}{2}}}{\Gamma(\frac{n+1}{2})}} \end{aligned}$$

Anti-de Sitter Space (AdS):

* AdS_D : ~~has~~ spherically symmetric metric

Derive the AdS_D -Schwarzschild metric

$$S = \frac{1}{2\kappa^2} \int d^D x \sqrt{g} (R - \Lambda)$$

where $\Lambda \equiv - (D-1)(D-2)/L^2$, L :

Motivation: AdS Supergravity

The covariant derivative D_μ on AdS_D acts differently on spinors. $D_\mu \psi_p = \partial_\mu \psi_p + \frac{1}{4} \omega_{\mu ab} \gamma^{ab} \psi_p$

Since we will obtain an extra

$\delta \epsilon$ from the Action which we "throw" into the covariant Der. of ψ_p to "counter" it (ROUGHLY):

$$\delta \epsilon = \frac{1}{2} e (\bar{\epsilon} \gamma^\mu \psi_p)$$

$$\therefore \Delta \delta \epsilon = (\Delta \frac{1}{2}) e (\bar{\epsilon} \gamma^\mu \psi_p)$$

$$\Rightarrow \tilde{D}_\mu \psi_p = D_\mu \psi_p + \hat{\Delta}_\mu \psi_p$$

$$\Rightarrow \hat{\Delta}_\mu \approx \frac{\Delta}{2} e \gamma_\mu$$

Thus, in AdS_D : $\epsilon(x) \in \Lambda^1(M)$

$$\boxed{\tilde{D}_\mu \epsilon \equiv (D_\mu - \frac{1}{2L} \gamma_\mu) \epsilon = (\partial_\mu + \frac{1}{4} \omega_{\mu ab} \gamma^{ab} - \frac{1}{2L} \gamma_\mu) \epsilon}$$

$$\begin{aligned}
[D_m, D_n] &= [D_m, D_n] + [\tilde{Y}_m, \tilde{Y}_n] e \\
&= [D_m, D_n] + [\tilde{Y}_m, \tilde{Y}_n] e \\
&= \frac{1}{4} R_{mnab} \gamma^{ab} e + \frac{1}{(2L)^2} e_m^a e_n^b \gamma_{ab} - e_m^b e_n^a \gamma_{ba} \\
&= \frac{1}{4} (R_{mnab} \gamma^{ab} + \frac{1}{L^2} (e_m^a e_n^b - e_n^a e_m^b) \gamma^{ab}) e \\
&= \frac{1}{4} (R_{mnab} + \frac{1}{L^2} (e_m^a e_n^b - e_n^a e_m^b) \gamma^{ab}) e \\
&= \frac{1}{4} \hat{R}_{mnab} \gamma^{ab} e
\end{aligned}$$

AdS₅ Curvature:

$$\hat{R}_{mnab} = R_{mnab} + \frac{1}{L^2} (e_m^a e_n^b - e_n^a e_m^b)$$

Ricci: $\hat{R}_{ma} = R_{mnab} e^{bn} = R_{ma} + \frac{1}{L^2} (e_m^a D - e_{am})$

$$= R_{ma} + \frac{D-1}{L^2} e_{am}$$

$$\hat{R} = \hat{R}_{ma} e^{am} = R + \frac{D(D-1)}{L^2}$$

• As we shall see, AdS₅ is a solution of the equation $\hat{R}_{mn} = 0$

• Obtaining AdS SUGRA from M^D :

$$D_m \rightarrow \nabla_m \text{ and } R \rightarrow \hat{R}_{D-1}$$

$$\hat{R} = \frac{(D-1)(D-2)}{L^2} + R$$

$$\therefore \frac{1}{2\kappa^2} \int d^D x e (R - \Lambda) = \frac{1}{2\kappa^2} \int d^D x e \hat{R} = \frac{1}{2\kappa^2} \int d^D x e (R + \frac{(D-1)(D-2)}{L^2})$$

$$\Rightarrow \Lambda = -\frac{(D-1)(D-2)}{L^2}$$

gravitational

Claim: The ^VE.O.M. is given by

$$\hat{R}_{mn} = 0 = R_{mn} + \frac{(D-2)}{L^2} g_{mn}$$

Proof:

$$\begin{aligned}
eSR &= \delta(e^{bn} R_{mnab} e^{ma}) \quad \leftarrow SR \sim D_m \Gamma \dots \\
&= \delta e^{bn} R e^{ma} + e^{bn} \delta R e^{ma} + e^{bn} R \delta e^{ma} \\
&= \delta e^{bn} R e^{ma} + e^{bn} \delta R e^{ma} + e^{bn} R \delta e^{ma}
\end{aligned}$$

$$= \delta e^{\ln R} R_{mnab} e^a + e^a R_{mnb} \delta e^{\ln R}$$

$$= 2 e^{\ln R} R_{mnab} \delta e^a = 2 R_{na} \delta e^a e$$

$$\delta e = e e^a \delta e_a = -e e_{an} \delta e^a$$

$$e e_{an} \delta e^a (R-1) + e 2 R_{an} \delta e^a = 0$$

$$2 R_{an} - e_{an} (R-1) = 0$$

$$2 R_{mn} - g_{mn} (R-1) = 0$$

$$2R - D(R-1) = 0$$

$$(D-2)R = D \Rightarrow R = \frac{D}{D-2}$$

$$\Rightarrow 2R_{mn} - g_{mn} \Lambda \left[\frac{D}{(D-2)} - \frac{(D-2)}{(D-2)} \right] = 0$$

$$R_{mn} - g_{mn} \Lambda \left(\frac{1}{D-2} \right) = 0$$

$$R_{mn} = g_{mn} \left(\frac{1}{D-2} \right) \left(\frac{D-2}{L^2} \right)$$

$$\therefore R_{mn} = - \frac{(D-2)}{L^2} g_{mn} \quad R = - \frac{(D-2)D}{L^2}$$

Note, that AdS is maximally symmetric,

$$\text{so } R_{mnpq} = \left[\frac{R}{D(D-2)} \right] (g_{mp} g_{nq} - g_{mq} g_{np})$$

$$\text{For, AdS}_D: R_{mnpq} = \left(- \frac{1}{L^2} \right) (g_{mp} g_{nq} - g_{mq} g_{np})$$

Any metric which satisfies are called

$$R_{mn} = K(D-2) g_{mn}$$

$$R = K D(D-2)$$

(Einstein
Metrics)

* A maximally symmetric spacetime of dim D can be embedded as a hyperboloid in flat spacetime of dim D+2.

g_{mn} : Induced metric

$\uparrow$ $D+2$

Ad

AdS_D Obtained from its embedding in $\mathbb{R}^{D+1}$

The embedding space $\mathbb{R}^{D+1} = \mathbb{R}^{2, D-2}$

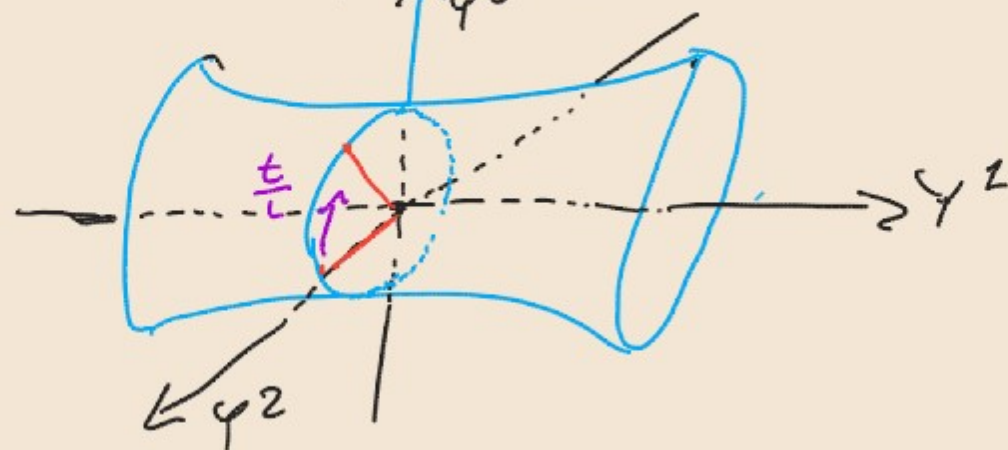
$\eta_{AB} = (-++ \dots + -)$ w/ Cartesian coords $Y^A, A=0, \dots, D$

The hyperboloid that defines the AdS_D

Surface:

$$f(Y) \equiv Y^A \eta_{AB} Y^B = -(Y^0)^2 + \sum_{i=1}^{D-1} (Y^i)^2 - (Y^D)^2 = -L^2$$

* For $D=2$, $A=0, 1, 2$



Isometry group of AdS_D in $\mathbb{R}^{D+1}$ is

$$SO(D-2, 2) \ni \Lambda^A_B \Leftrightarrow \Lambda^D_A \eta_{DC} \Lambda^C_B = \eta_{AB}$$

$$Y'^A = \Lambda^A_B Y^B$$

• As in the Lorentz case (NOT Poincaré, which includes translations):

$$U_L(L) = e^{-\frac{i}{2} \lambda^{ab} L_{ab}}, \quad L_{ab} = x_a \partial_b - x_b \partial_a$$

("Angular-momentum" operator which is induced by rotations.

$$L_{AB} = Y_A \frac{\partial}{\partial Y_B} - Y_B \frac{\partial}{\partial Y_A}$$

Claim: At the level of algebras, the invariance of the surface:

$$U_L f(Y) = f(Y') \Leftrightarrow L_{AB} f(Y) = 0$$

Proof: $L_{AB} f(Y) = (Y_A \partial_B - Y_B \partial_A)(Y^C M_{CD} Y^D)$

$$2Y_A M_{BD} Y^D - Y_B 2M_{AD} Y^D$$

$$2Y_A Y_B - 2Y_B Y_A = 0$$

Homogeneous space: Group action is transitive i.e. for any two pts $p, q \in \text{AdS}_D$ $\exists$

$$g \in \text{SO}(D-1, 2) \text{ s.t.}$$

$$p' = \sigma(g, p), \quad \sigma: G \times \text{AdS}_D \rightarrow \text{AdS}_D$$

* Maximal compact subgroup of $\text{SO}(D-1, 2)$

$$\text{SO}(D-1) \times \text{SO}(2)$$

Rotates Y^i
($i = 1, \dots, D-1$)

Rotates in the
O-D plane

Claim: Y^A can be brought to the point $(L, 0, \dots, 0)$

Proof: $Y^A = (Y^0, Y^1, \dots, Y^{D-1}, Y^D)$

Require the matrix

$$M \begin{pmatrix} Y^1 \\ \vdots \\ Y^{D-1} \end{pmatrix} = \begin{pmatrix} Y^{D-1} \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

Which we know exists, this is simply a Boost of a massless particle in $\mathbb{R}^{D-1}$

$$\therefore Y^A \xrightarrow{M} (Y^0, Y^{D-1}, \dots, Y^D)$$

$$\tilde{M} \in SO(2) \cong U(1)$$

$$\tilde{M} \begin{pmatrix} \gamma^0 \\ \gamma^D \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} \gamma^0 \\ \gamma^D \end{pmatrix}$$

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a^2 + b^2 & ac + bd \\ ac + db & c^2 + d^2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$c^2 + d^2 = 1, \quad ac + bd = 0, \quad a^2 + b^2 = 1$$

$$ad - cb = 1$$

$$\bullet \quad b = \sqrt{1 - a^2}, \quad d = \frac{1 + cb}{a}$$

$$c = \sqrt{1 - d^2}$$

$$\gamma^0 c + d \gamma^D = 0$$

$$\gamma^{02} c^2 = d^2 \gamma^{D2}$$

$$(\gamma^0)^2 (1 - d^2) = (\gamma^D)^2 d^2$$

$$(\gamma^0)^2 = d^2 ((\gamma^0)^2 + (\gamma^D)^2)$$

$$\Rightarrow d^2 = (\gamma^0)^2 / ((\gamma^0)^2 + (\gamma^D)^2)$$

$$\Rightarrow \tilde{M} = \begin{pmatrix} a & b \\ \sqrt{1-d^2} & d \end{pmatrix} \uparrow$$

$$\gamma^A \xrightarrow{\tilde{M}} (\gamma^0, \gamma^{1,2}, \dots, \gamma^D) \xrightarrow{\tilde{M}} (\gamma^{0'}, \gamma^{1'}, 0, \dots, 0)$$

Now, exactly as we just did for (γ^0, γ^D)
we act w/ $SO(2)$ to obtain

$$\gamma^A \xrightarrow{\tilde{M}} (\gamma^0, \gamma^{1,2}, \dots, 0, \gamma^D) \xrightarrow{\tilde{M}} (\gamma^{0'}, \gamma^{1'}, 0) \xrightarrow{\tilde{M}} (L, 0, \dots, 0)$$

* Different coordinate systems of AdS_D

x^m : $m = 0, 1, \dots, D-2$ "Intrinsic coordinates"

• These intrinsic coordinates generate different
-morphisms which are Killing vectors.

$$Y^i \equiv r \bar{X}^i \quad w/ \quad \sum_{i=1}^{D-2} (\bar{X}^i)^2 = 1$$

$$Y^0 = \sqrt{L^2 + r^2} \sin\left(\frac{t}{L}\right), \quad Y^D = \sqrt{L^2 + r^2} \cos\left(\frac{t}{L}\right)$$

$\bar{X}^i$: parametrizes the unit sphere S^{D-2}

$\hookrightarrow$ further parametrized by hyperspherical coordinates.

$$ds^2 = \eta_{AB} dY^A dY^B = -(dY^0)^2 + \sum_{i=1}^{D-2} (dY^i)^2 - (dY^D)^2$$

$$dY^0 = \left(\frac{dr}{\sqrt{L^2 + r^2}}\right) r \sin\left(\frac{t}{L}\right) + \sqrt{L^2 + r^2} \cos\left(\frac{t}{L}\right) \frac{dt}{L}$$

$$(dY^0)^2 = \left(\frac{r^2}{L^2 + r^2}\right) \sin^2\left(\frac{t}{L}\right) dr^2 + \left(\frac{L^2 + r^2}{L^2}\right) \cos^2\left(\frac{t}{L}\right) (dt)^2$$

$$+ 2 \sin\left(\frac{t}{L}\right) \cos\left(\frac{t}{L}\right) \left(\frac{dt dr}{L}\right) r$$

$$(dY^D) = \left(\frac{dr}{\sqrt{L^2 + r^2}}\right) r \cos\left(\frac{t}{L}\right) - \sqrt{L^2 + r^2} \sin\left(\frac{t}{L}\right) \frac{dt}{L}$$

$$(dY^D)^2 = \left(\frac{r^2}{L^2 + r^2}\right) \cos^2\left(\frac{t}{L}\right) dr^2 + \left(\frac{L^2 + r^2}{L^2}\right) \sin^2\left(\frac{t}{L}\right) (dt)^2$$

$$- 2 \left(\frac{r}{L}\right) \sin\left(\frac{t}{L}\right) \cos\left(\frac{t}{L}\right) dt dr$$

$$- \left((dY^0)^2 + (dY^D)^2 \right) = - \left[\frac{r^2}{L^2 + r^2} dr^2 + \left(1 + \left(\frac{r}{L}\right)^2\right) (dt)^2 \right]$$

$$dY^i = dr \bar{X}^i + r d\bar{X}^i$$

$$(dY^i)^2 = (dr)^2 (\bar{X}^i)^2 + 2r \bar{X}^i dr d\bar{X}^i + r^2 (d\bar{X}^i)^2$$

$$\sum_{i=1}^{D-2} (dY^i)^2 = (dr)^2 + r^2 \sum_{i=1}^{D-2} (d\bar{X}^i)^2$$

$$= (dr)^2 + r^2 d\Omega_{(D-2)}^2$$

$$ds^2 = (dr)^2 \left[1 - \frac{r^2}{L^2 + r^2} \right] - \left(1 + \left(\frac{r}{L}\right)^2 \right) (dt)^2$$

$$+ r^2 d\Omega_{(D-2)}^2$$

$$= \left(1 + \left(\frac{r}{L}\right)^2 \right)^{-1} (dr)^2 - \left(1 + \left(\frac{r}{L}\right)^2 \right) (dt)^2$$

$$ds^2 = -(1 + (\frac{r}{L})^2) dt^2 + (1 - (\frac{r}{L})^2) dr^2 + r^2 d\Omega_{(D-2)}^2$$

(Global coordinate system)

$$0 \leq r < \infty, \theta^i, 0 \leq t < 2\pi L$$

* Note, "time" t is periodic, so AdS_3 contains closed timelike curves

→ For fixed r and $\bar{x}^i$ (i.e. θ^i)

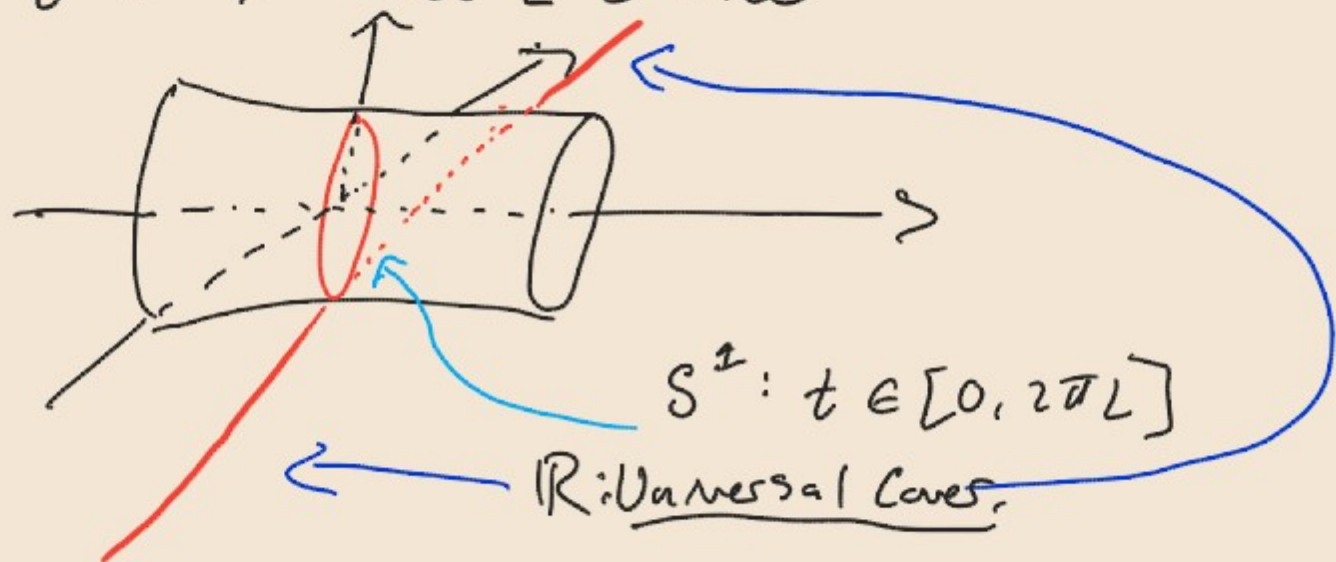
$$\begin{aligned} ds^2 &= -(1 + (\frac{r}{L})^2) dt^2 \\ ds^2 &= -d\tau^2 \quad \text{so} \\ \tau &= \sqrt{1 + (\frac{r}{L})^2} \int_0^{2\pi L} dt = 2\pi L \sqrt{1 + (\frac{r}{L})^2} \end{aligned}$$

? Not sure

• We have closed timelike curves.

* To have a physical theory we must move to the covering space:

Unwrap S^1 and analytically continue $t \rightarrow -\infty \leq t < \infty$



Aside: Universal Covering Space/Group:

Def: Let X and $\tilde{X}$ be connected top. spaces. The pair $(\tilde{X}, p)$ [or simply $\tilde{X}$] is called a covering space of X if $\exists$ a continuous map $p: \tilde{X} \rightarrow X$ s.t.

(ii) $\forall x \in X \exists$ a connected open set $U \subset X$ w/ $x \in U$ s.t. $p^{-1}(U) = \bigsqcup_i V_i$ s.t. $V_i \subset \tilde{X}$ and each V_i is mapped homeomorphically onto U by p .

Def: If $\tilde{X}$ is simply connected $(\tilde{X}, p)$ is called the universal covering space of X .

* $\boxed{\mathbb{R} \xrightarrow{p} S^1}$, $\mathbb{R}$ is the universal covering space of S^1 . $p: \mathbb{R} \rightarrow U(1)$, $p: x \mapsto e^{2\pi i x}$

$$\text{Ad}S_D \cong S^{D-2} \times \mathbb{R} \times S^1$$

$$U(\text{Ad}S_D) \rightarrow \text{Ad}S_D$$

$$U\text{Ad}S_D \cong S^{D-2} \times \mathbb{R}^2$$

$\Rightarrow r=0$ is not distinguished and there is therefore no set origin, since $\text{Ad}S_D$ is homogeneous.

$\Rightarrow$ Any point on $\text{Ad}S_D$ can be made the origin!

* The time translation $t \rightarrow t+a$, $a \in \mathbb{R}$ is an isometry of the metric. This is generated by $\partial/\partial t$:

$$U_a(t) = (e^{a \frac{\partial}{\partial t}})(t) = t + a \frac{\partial}{\partial t} t = t + a$$

$\Rightarrow \partial/\partial t$ is a Killing vector.

Note: $L_{\partial/\partial t} = \gamma_0 \frac{\partial}{\partial \gamma_0} - \dot{\gamma}_0 \frac{\partial}{\partial \dot{\gamma}_0}$

$$\frac{\partial}{\partial \gamma_0} = \frac{\partial r}{\partial \gamma_0} \frac{\partial}{\partial r} + \frac{\partial \dot{\gamma}_0}{\partial \gamma_0} \frac{\partial}{\partial \dot{\gamma}_0}$$

$$\frac{(\dot{\gamma}_0)^2}{c^2} = 1^2 + r^2 \Rightarrow 2r \frac{\partial r}{\partial \gamma_0} = \frac{2\dot{\gamma}_0}{c^2}$$

$$\frac{\partial}{\partial r} = \frac{1}{r} \frac{\partial}{\partial \sigma}$$

$$1 = \sqrt{\cos(\frac{t}{L})} \left(\frac{\partial}{\partial r} \right) \frac{1}{L}$$

$$L_{00} = L \frac{\partial}{\partial t}$$

$$\cosh^2 - \sinh^2 = 1$$

Other coordinate systems: change the radial coordinate $r \rightarrow y$

$$\cosh\left(\frac{y}{L}\right) := \sqrt{1 + (r/L)^2} \quad (\cosh^2 - 1)L^2 = r^2$$

$$\cosh^2 \sinh^2 dy^2 = \sinh^2 dr^2 \Leftrightarrow dr^2 = \cosh^2 dy^2$$

$$(1 + (\frac{r}{L})^2)^{-1} = \cosh^{-2}$$

$\Rightarrow$

$$0 \leq y < \infty$$

$$ds^2 = -\cosh^2\left(\frac{y}{L}\right) dt^2 + dy^2 + \sinh^2\left(\frac{y}{L}\right) L^2 d\Omega_{(p-2)}^2$$

Now, define $\cosh\left(\frac{y}{L}\right) =: \frac{1}{\cos(\rho)}$, $0 \leq \rho \leq \pi/2$
and Scaling $t = L\tau$

$$\Rightarrow ds^2 = \left(\frac{L^2}{\cos^2(\rho)}\right) [-d\tau^2 + (d\rho^2 + \sinh^2(\rho) d\Omega_{(p-2)}^2)]$$

AdS_p w/ conformal factor $\frac{L^2}{\cos^2(\rho)}$ w/ the conformal metric $\tilde{ds}^2$ which is locally $\cong \mathbb{R} \times S^{p-1}$, $-\infty < \tau < \infty$

$\sim$ Require $0 \leq \rho < \pi$ to cover all of AdS ,
so $0 \leq \rho < \pi/2$ only covers the upper hemisphere

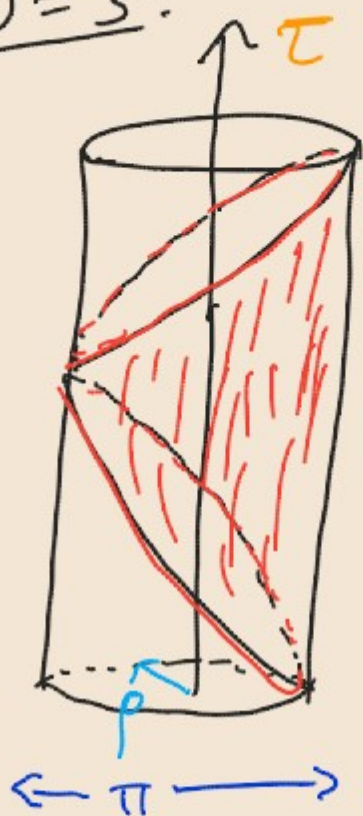
$\therefore AdS_p \cong_c$ (Einstein static Universe)⁺

In words, AdS_p is conformal to the upper half of an Einstein static universe.

Global view of AdS:

→ (To be expanded on)

$D=3$:



"Radius": $\rho_0 = \pi/2$:

- Does not make sense since conformally $ds^2 \rightarrow \infty$ as $\rho_0 \rightarrow \pi/2$
- Boundary at " ∞ " distance away

For higher dim. cases ($D > 3$) each interior pt. represents an entire S^{D-3} .

* Information from spatial ∞ reaches the origin in finite time.

Poincaré-patch: Patch which is conformal to half of AdS^D .

$$y^0 \equiv \ln x^0$$

$$y^i \equiv \ln x^i, \quad i = 1, \dots, D-2$$

$$y^{D-1} \equiv \frac{1}{2\alpha} (-1 + \alpha^2 (L^2 - x^2)), \quad y^D \equiv \frac{1}{2\alpha} (1 + \alpha^2 (L^2 + x^2))$$

$$x^2 \equiv -(x^0)^2 + \sum_{i=1}^{D-2} (x^i)^2$$

$$x^0 = t, \quad -\infty < t < \infty \quad | \quad x^i: -\infty < x^i < \infty$$

$$0 < \alpha < \infty$$

Note:

$$y^0, y^i, y^{D-1}, y^D$$

$$\begin{aligned}
& \rightarrow L^2 u^2 (x^0)^2 + L^2 u^2 \sum_i (x^i)^2 + \frac{1}{4u^2} (-1 + u^2(L^2 - x^2))^2 \\
& \quad - \frac{1}{4u^2} (1 + u^2(L^2 + x^2))^2 \\
& = L^2 u^2 x^2 + \frac{1}{4u^2} [1 - 2u^2(L^2 - x^2) + u^4(L^2 - x^2)^2] \\
& \quad - \frac{1}{4u^2} [1 + 2u^2(L^2 + x^2) + u^4(L^2 + x^2)^2] \\
& = L^2 u^2 x^2 - L^2 + \frac{u^2}{4} (L^4 - 2x^2 L^2 + x^4) \\
& \quad - \frac{u^2}{4} (L^4 + 2x^2 L^2 + x^4) \\
& = L^2 u^2 x^2 - L^2 - u^2 x^2 L^2 = -L^2 \quad \checkmark
\end{aligned}$$

Now,

$$dy^0 = L du x^0 + L u dx^0$$

$$(dy^0)^2 = L^2 (x^0)^2 (du)^2 + 2L^2 u x^0 du dx^0 + L^2 u^2 (dx^0)^2$$

$$(dy^i)^2 = L^2 (x^i)^2 du^2 + 2L^2 u x^i du dx^i + L^2 u^2 (dx^i)^2$$

$$\begin{aligned}
dy^{D-1} &= -\frac{1}{2u^2} du (-1 + u^2(L^2 - x^2)) + \frac{1}{2} \sum_i x^i dx^i \\
& \quad + \frac{1}{2u} (2u du (L^2 - x^2) + u^2 (-1) (-x^0 dx^0)) \\
& = \left\{ \frac{1}{2u^2} [-1 + u^2(L^2 - x^2)] + L^2 - x^2 \right\} du \\
& \quad - \frac{u}{2} [-2x^0 dx^0 + 2 \sum_i x^i dx^i] \\
& = \left[\frac{1}{2u^2} - \frac{L^2}{2} + \left(\frac{x}{u}\right)^2 \frac{1}{2} \right] du - \frac{u}{2} [-2x^0 dx^0 + 2 \sum_i x^i dx^i] \\
& = \left[\frac{1}{2u^2} (1 + x^2) - \frac{L^2}{2} \right] du - u \left\{ -x^0 dx^0 + \sum_i x^i dx^i \right\} \\
(dy^{D-1})^2 &= \left(\frac{1}{2u^2} (1 + x^2) - \frac{L^2}{2} \right)^2 du^2 + \frac{u^2}{4} (\quad)^2 \\
& \quad - 2u [\quad]
\end{aligned}$$

$$(dy^0)^2 + \sum_{i=1}^{D-2} (dy^i)^2$$

$$\begin{aligned}
& = -L^2 du^2 (x^0)^2 - 2L^2 u x^0 du dx^0 - L^2 u^2 (dx^0)^2 \\
& \quad + L^2 du^2 \sum_{i=1}^{D-2} (x^i)^2 + 2L^2 u du \sum_i x^i dx^i \\
& \quad + L^2 u^2 \sum_i (dx^i)^2 \\
& = L^2 du^2 x^2 + L^2 u^2 \left[-(dx^0)^2 + \sum_{i=1}^{D-2} (dx^i)^2 \right]
\end{aligned}$$

... we find

$$ds^2 = L^2 \left[\frac{du^2}{u^2} + u^2 \left(-(dx^0)^2 + \sum_i (dx^i)^2 \right) \right]$$

(Poincaré Patch Metric)

Define: $z := \frac{1}{u} \Rightarrow du = -\frac{1}{z^2} dz$

$$ds^2 = L^2 \left[\frac{1}{z^2} dz^2 + \frac{1}{z^2} \left(-(dx^0)^2 + \sum_i (dx^i)^2 \right) \right]$$

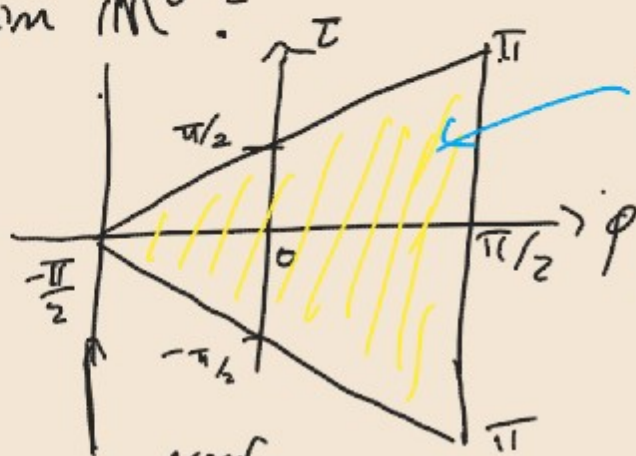
$\Rightarrow$

$$ds^2 = \frac{L^2}{z^2} \left[dz^2 - (dx^0)^2 + \sum_i (dx^i)^2 \right]$$

so it is conformal to M^D w.r. (x^0, x^i, z)

- In the limit $z \rightarrow 0$ one reaches the Boundary of the Poincaré Patch. Which is the conf. comp. of $(D-1)$ -dim M^{D-2} .

AdS₂:



Poincaré Patch

Since $AdS_2 \cong M^2$: $ds^2 = -dx^0^2 + dz^2$

* Compare Local and Global Coordinates

$$\begin{cases} \sqrt{L^2 + r^2} \sin(\frac{\phi}{L}) = L u x^0 \\ \sqrt{L^2 + r^2} \cos(\frac{\phi}{L}) = \frac{1}{2u} (1 + u^2 (L^2 + x^2)) \\ r \bar{x}^i = \frac{1}{2u} (-1 + u^2 (L^2 - x^2)) \end{cases} \begin{cases} x^0 = t \\ x^2 = t^2 \\ \bar{x}^i = 1 \end{cases}$$

$$\begin{aligned} \bullet \left(L^2 \frac{1}{\cos(\phi)} \right) &= \sqrt{L^2 + r^2} \\ \Rightarrow r &= \frac{L \sin(\phi)}{\cos(\phi)} \end{aligned}$$

$$\therefore \left(\frac{L^2}{r^2} \right) = \left(\frac{1}{\sin^2(\phi)} \right) = \frac{1}{\sin^2(\phi)}$$

$$\frac{\sinh(\tau)}{\cos(\rho)} = 21 \tau$$

$$(i) L^2 \frac{\cos(\tau)}{\cos(\rho)} = \frac{1}{2u} (1 + u^2 (L^2 + t^2))$$

$$(ii) L \frac{\sinh(\tau)}{\cos(\rho)} = \frac{1}{2u} (-1 + u^2 (L^2 - t^2))$$

$$V_{Sch} \quad z = \frac{1}{u}$$

$$t = \frac{L \sinh(\tau)}{\cos(\tau) + \sinh(\rho)}$$

$$z = L \frac{\cos(\rho)}{\cos(\tau) + \sinh(\rho)}$$

* Note:

(a) all observers which follow curves of constant z reach the same Boundary point as $t \rightarrow \infty$:

$$\cos(\tau) + \sinh(\rho) = 0$$

$$\Rightarrow (\tau = \pi, \rho = \pi/2)$$



(b) Lines for $z = \infty$: Horizon

Since observers cannot receive info from the region above (nor send info to region below) the Horizon.

$$\partial \text{AdS}_2 = (-\infty < t < \infty, \rho = \pi/2) \cup (t=0, \rho = -\pi/2)$$

$$\frac{t}{z} = \frac{\sinh(\tau)}{\cos(\rho)}$$

$$\text{So } z \rightarrow \infty \text{ implies } \rho = \pm \pi/2$$



• Lines of constant z are time-like but non-geodesic
 - Time-like geodesics are periodic on the Hyperboloid.

and best described by global coordinates.

Symmetries of the Poincaré Patch:

* Define $x^\sigma \equiv (x^0, x^i)$ $(D-1)$ -components
which is contracted by $\eta_{\sigma\lambda} := (- + \dots +)$

(i) (Translations) $(D-1)$ -coords x^σ
(ii) $(D-1)(D-2)/2$ distinct Lorentz transf. of
 x^σ : $SO(D-1)$

(iii) (Weyl) Scale transf.

$$\delta x^0 = p x^0, \quad \delta x^i = p x^i, \quad \delta z = p z$$

$$p \ll 1,$$

(iv) (SCTs) $(D-1)$ distinct SCTs

$$\delta x^\sigma = c^\sigma (x \cdot x + z^2) - 2 c \cdot x x^\sigma$$

$$\delta z = -2 c \cdot x z,$$

c^σ constant inf. parameter vector ($c^z = 0$)

- Note, only inv. for some (such as translations)
due to the fact that $ds^2 = \frac{1}{z^2}$
so $z \rightarrow z + a$ is not a symmetry.

$$\begin{aligned} \text{Total} &= (D-1) + \frac{(D-1)(D-2)}{2} + 1 + (D-1) \\ &= \frac{1}{2}(D-1) + \frac{(D-1)(D-2)}{2} + \frac{2}{2} \\ &= \frac{1}{2} [4(D-1) + 2 + (D-1)(D-2)] \\ &= \frac{1}{2} [(D-1)[D+2] + 2] \\ &= \frac{D(D+1)}{2} \end{aligned}$$

$$\dim(SO(D-1, 2)) = \frac{D(D+1)}{2}$$

$D-1$

equal to the isometry group of AdS_D .
 $\mathbb{M}^{D-2} = 2\text{AdS}_D$ [Poincaré Patch]

Claim: Vector fields for the above transf. are Killing

Proof:

$$(i) \quad x^\sigma \rightarrow x^\sigma + a^\sigma = e^{a^\sigma \partial_\sigma} x^\sigma = x^\sigma + a^\sigma \partial_\sigma x^\sigma$$

$$\text{Trivially } \nabla_\mu a_\sigma = 0$$

$$(ii) \quad \delta x^\sigma = \Lambda^\sigma_\lambda x^\lambda$$

$$\nabla_\mu (\Lambda^\sigma_\lambda x^\lambda) = g_{\mu\alpha} \nabla_\mu \Lambda^\sigma_\lambda x^\lambda + \Lambda^\sigma_\mu g_{\alpha\nu} x^\nu$$

$$\nabla_\mu (\Lambda^\sigma_\lambda x^\lambda) = 0$$

(iii) Trivial, But show that the Killing vec. is equal to $L_D(D-1)$.

Killing vect. of above is $(\partial x^\mu \partial_\mu)$

$$\frac{\partial}{\partial y^D} = \frac{\partial z}{\partial y^D} \frac{\partial}{\partial z} + \frac{\partial x^0}{\partial y^D} \frac{\partial}{\partial x^0} + \frac{\partial x^i}{\partial y^D} \frac{\partial}{\partial x^i}$$

$$y^D = \frac{z}{2} \left(1 + \frac{1}{z^2} (L^2 - (x^0)^2 + \sum_i (x^i)^2) \right)$$

$$2y^D = \left(\frac{z^2}{2} + (L^2 - (x^0)^2 + \sum_i (x^i)^2) \right)$$

$$z^2 - (2y^D z) + 2(L^2 - (x^0)^2 + \sum_i (x^i)^2) = 0$$

$$\frac{2y^D \pm 2\sqrt{(y^D)^2 - (L^2)}}{2} = z$$

$$z = y^D \pm \sqrt{(y^D)^2 - (L^2)}$$

$$\frac{\partial z}{\partial y^D} = 1 + \frac{y^D}{\sqrt{(y^D)^2 - (L^2)}}$$

Spacetime metrics of Spherical symmetry:

• Any spherically symmetric vacuum metric possesses a timelike Killing vector, ds^2

is called stationary. ds^2 is static if it possesses a timelike Killing vector which is orthogonal to a family of hypersurfaces.

$$ds^2 = -e^{2A(r)} dt^2 + e^{2B(r)} dr^2 + r^2 d\Omega_{(n-2)}^2$$

• Recall $d\Omega_{(n)}^2 = \delta_{ab} \bar{e}^a \bar{e}^b$ and the Cartan structure eqs:

$$de^a + \omega^a_b \wedge e^b = T^a \quad \left| \quad \rho^{ab} = \frac{1}{2} R_{mn}^{ab} dx^m dx^n \right.$$

$$d\omega^{ab} + \omega^a_c \wedge \omega^{cb} = \rho^{ab}$$

• Torsion free: $T^a = 0$

$$de^a = -\omega^a_b \wedge e^b$$

We rewrite the metric

$$ds^2 = -(e^t)^2 + (e^r)^2 + r^2 \bar{e}^i \bar{e}_i$$

$$e^t \equiv e^{A(r)} dt$$

$$e^i \equiv r \bar{e}^i$$

$$e^r \equiv e^{B(r)} dr$$

$$ds^2 = -(e^t)^2 + e^r \cdot e^r + \sum e^i \cdot e^i$$

$$\begin{aligned} de^t &= e^t \frac{dA}{dr} dr dt = \frac{dA}{dr} \bar{e}^r e^r \wedge e^t \\ &= \omega^t_r \wedge e^r = \kappa_{cd} \omega^{td} \wedge e^c \\ &= -\omega^{tr} \wedge e^r = \omega^{ti} \wedge e^i \end{aligned}$$

$$\therefore de^t = A' \bar{e}^r e^r \wedge e^t = -\omega^{tr} \wedge e^r - \omega^{ti} \wedge e^i$$

$$de^r = 0 = \omega^{rd} \kappa_{cd} \wedge e^d = -\omega^{rt} \wedge e^t + \omega^{ri} \wedge e^i$$

$$de^i = dr \wedge \bar{e}^i + r d\bar{e}^i$$

$$= +\omega^{it} \wedge e^t - \omega^{ir} \wedge e^r - \omega^{ij} \wedge e^j$$

$$de^i = dr \wedge \bar{e}^i - \bar{\omega}^{ij} \wedge e^j = -\omega^{ij} \wedge e^j - \omega^{ir} \wedge e^r + \omega^{it} \wedge e^t$$

$$= e^{-\beta(r)} e^r \wedge \bar{e}^i - \bar{\omega}^{ij} \wedge e^j = \dots$$

$$= \frac{e^{-\beta}}{r} e^r \wedge e^i - \bar{\omega}^{ij} \wedge e^j = -\omega^{ij} \wedge e^j - \omega^{ir} \wedge e^r + \omega^{it} \wedge e^t$$

$$\Rightarrow \bar{\omega}^{ij} = \omega^{ij}$$

$$(e^{-\beta}/r) e^r \wedge e^i = -\omega^{ir} \wedge e^r + \omega^{it} \wedge e^t$$

$$-\frac{e^{-\beta}}{r} e^i \wedge e^r = -\omega^{ir} \wedge e^r + \omega^{it} \wedge e^t$$

$$\Rightarrow \omega^{ir} = \frac{e^{-\beta}}{r} e^i$$

Since $e^t \neq 0$, $\omega^{it} = 0$

$$\Rightarrow A' e^{-\beta} e^r \wedge e^t = -\omega^{tr} \wedge e^r$$

$$\omega^{tr} = A' e^{-\beta} e^t$$

All together we find:

$$e^t = e^A dt, \quad e^r = e^B dr, \quad e^i = r \bar{e}^i$$

$$\omega^{tr} = A' e^{-\beta} e^t, \quad \omega^{ti} = 0$$

$$\omega^{ir} = (e^{-\beta}/r) e^i, \quad \omega^{ij} = \bar{\omega}^{ij}$$

We can now calc. the curvature 2-form

$$\rho^{ab} = \frac{1}{2} R_{\mu\nu}^{\quad ab} dx^\mu \wedge dx^\nu$$

$$(i) \underline{\omega^{tr}} = (A'' e^{-\beta} - A' B' e^{-2\beta}) dr \wedge e^t$$

$$+ A'' e^A dr \wedge e^t$$

$$= (A'' e^{-2\beta} - A' B' e^{-2\beta} + A'' e^{-\beta}) e^r \wedge e^t$$

$$+ A' e^{-2\beta} e^r \wedge e^t$$

$$= (A'' e^{-2\beta} - A' B' e^{-2\beta} + A'' e^{-\beta}) e^r \wedge e^t$$

$$\bullet \omega^{tc} \wedge \omega_c^r = \omega^{tr} \wedge \omega^r_r + \omega^{ti} \wedge \omega^i_r = 0$$

$\therefore$

$$\rho^{tr} = (A'' - A' B' + (A')^2) e^{-2\beta} e^r \wedge e^t$$

$$\rho^{tr} = -(A'' - A' B' + (A')^2) e^{-2\beta} e^t \wedge e^r$$

$$(i) \rho^{ti} = d\omega^{ti} + \omega^{t\alpha} \wedge \omega_{\alpha}{}^i$$

$$d\omega^{ti} = 0$$

$$\omega^{tr} \wedge \omega_r{}^i + \omega^{t\alpha} \wedge \omega_{\alpha}{}^i$$

$$= (A' e^{-\beta} e^t) \wedge (-e^{-\beta}/r) e^i$$

$$= -\frac{A'}{r} e^{-2\beta} e^t \wedge e^i$$

$$\boxed{\rho^{ti} = -\frac{A'}{r} e^{-2\beta} e^t \wedge e^i}$$

$$(ii) \rho^{ir} = d\omega^{ir} + \omega^{i\alpha} \wedge \omega_{\alpha}{}^r$$

$$d\omega^{ir} = d(e^{-\beta} \bar{e}^i) = -\beta' e^{-\beta} e^r \wedge \bar{e}^i$$

$$+ \bar{e}^i d\bar{e}^i$$

$$\text{but } d\bar{e}^i = \frac{de^i - dr \wedge \bar{e}^i}{r}$$

$$d\omega^{ir} = -\beta' e^{-2\beta} e^r \wedge \bar{e}^i + \frac{\bar{e}^i}{r} de^i - \frac{dr \wedge \bar{e}^i}{r} e^{-\beta}$$

$$= -\frac{\beta'}{r} e^{-2\beta} e^r \wedge e^i + \dots$$

$$\rightarrow \omega^{i\alpha} \wedge \omega_{\alpha}{}^r = \omega^{ik} \wedge \omega_k{}^r + \omega^{it} \wedge \omega_t{}^r + \omega^{i\alpha} \wedge \omega_{\alpha}{}^r$$

$$=$$

$$\Rightarrow \boxed{\rho^{ir} = \frac{\beta'}{r} e^{-2\beta} e^r \wedge e^i}$$

$$(iv) \rho^{ij} = d\omega^{ij} + \omega^{i\alpha} \wedge \omega_{\alpha}{}^j$$

$$= d\omega^{ij} - \omega^{it} \wedge \omega_t{}^j + \omega^{ir} \wedge \omega_r{}^j + \omega^{ik} \wedge \omega_k{}^j$$

$$= d\omega^{ij} - (e^{-\beta}/r) e^i \wedge (e^{-\beta}/r) e^j + \omega^{ik} \wedge \omega_k{}^j$$

$$= \bar{\rho}^{ij} - \frac{e^{-2\beta}}{r^2} e^i \wedge e^j$$

$$\text{But } \bar{\rho}^{ij} = \bar{e}^i \wedge \bar{e}^j, \text{ so}$$

$$\rho^{ij} = \frac{1}{r^2} e^i \wedge e^j - \frac{e^{-2\beta}}{r^2} e^i \wedge e^j$$

$$\boxed{\rho^{ij} = (1 - e^{-2\beta}) \frac{1}{r^2} e^i \wedge e^j}$$

• All together we find

$$\begin{aligned} p_{ti} &= -\frac{A'}{r} e^{-2B} e^t \wedge e^i \\ p_{ri} &= \frac{B'}{r} e^{-2B} e^r \wedge e^i \\ p_{ij} &= (1 - e^{-2B}) \frac{1}{r^2} e^i \wedge e^j \end{aligned}$$

$$R^a_b = R^{ac}_{bc}, \quad p^{ab} = \frac{1}{2} R^{ab}_{cd} e^c \wedge e^d$$

so,

$$\bullet p^{tr} = \frac{1}{2} R^{tr}_{cd} e^c \wedge e^d = -(\quad) e^t \wedge e^r$$

$$\Rightarrow R^{tr}_{tr} = -(\quad) e^{-2B}$$

$$\bullet p^{ti} = -\frac{A'}{r} e^{-2B} e^t \wedge e^i = R^{ti}_{ti} e^t \wedge e^i$$

$$R^{ti}_{ti} = -\frac{A'}{r} e^{-2B}$$

$$\bullet p^{ri} = \left(\frac{B'}{r}\right) e^{-2B} e^r \wedge e^i = R^{ri}_{ri} e^r \wedge e^i$$

$$R^{ri}_{ri} = \frac{B'}{r} e^{-2B}$$

$$\bullet p^{ij} \Rightarrow R^{ij}_{ij} = (1 - e^{-2B}) \frac{1}{r^2}$$

$$R^a_b = R^{at}_{bt} + R^{ar}_{br} + R^{ak}_{bk} \quad \Delta R^a_b \frac{1}{2} + \dots R^{aD-2}_{bD-2}$$

$$\begin{aligned} R^t_t &= R^{tt}_{tt} + R^{tr}_{tr} + R^{tk}_{tk} \\ &= -(A'' - A'B' + (A')^2) e^{-2B} - \frac{A'}{r} e^{-2B} (D-2) \end{aligned}$$

$$\therefore \underline{R^t_t = -[A'' - A'B' + (A')^2 + (D-2) \frac{A'}{r}] e^{-2B}}$$

$$\begin{aligned} R^r_r &= R^{rt}_{rt} + \sum R^{ri}_{ri} \\ &= -[A'' - A'B' + (A')^2] e^{-2B} + (D-2) \frac{B'}{r} e^{-2B} \end{aligned}$$

$$R^r_r = -[A'' - A'B' + (A')^2 - (D-2) \frac{B'}{r}] e^{-2B} \quad (1 - e^{-2B}) \frac{1}{r^2}$$

$$\begin{aligned} R^i_j &= R^{it}_{it} + R^{ir}_{ir} + R^{ik}_{ik} \\ &= \left[-\frac{A'}{r} e^{-2B}\right] + \left[\frac{B'}{r} e^{-2B}\right] + (D-3) \left[\frac{1}{r^2} (1 - e^{-2B})\right] \end{aligned}$$

$$\therefore R_{ij} = \delta_{ij} \left[\frac{B' - A'}{r} e^{-2B} + (D-3) \left(1 - e^{-2B} \right) \frac{1}{r^2} \right]$$

All together:

$$R_t^t = - \left[A'' - A' B' + (A')^2 + (D-2) \frac{A'}{r} \right] e^{-2B}$$

$$R_r^r = - \left[A'' - A' B' + (A')^2 - (D-2) \frac{B'}{r} \right] e^{-2B}$$

$$R_{ij}^i = \delta_{ij} \left[(D-3) \left(1 - e^{-2B} \right) \frac{1}{r^2} + \frac{B' - A'}{r} e^{-2B} \right]$$

AdS_D - Schwarzschild Spacetime:

Recall AdS_D is a solution to

$$R_{\mu\nu} = - \frac{(D-2)}{L^2} g_{\mu\nu}$$

$$g^{\mu\nu} R_{\mu\nu} = R^\sigma_\sigma = - \frac{(D-2)}{L^2} \delta^\sigma_\sigma$$

$$\Rightarrow R^t_t = - \frac{(D-2)}{L^2}, \quad (\delta^i_i R^i_i) / (D-2) = - \frac{(D-2)}{L^2}$$

$$R^r_r = - \frac{(D-2)}{L^2}$$

$$\Leftrightarrow R^t_t = R^r_r \Leftrightarrow A' = -B'$$

$$A(r) + B(r) = f(\tau)$$

Rescale $f(\tau)$ s.t. $f(\tau) = 0$

$$\Leftrightarrow A(r) = -B(r)$$

$$R^i_i = (D-2) \left[(D-3) \left(1 - e^{-2B} \right) \frac{1}{r^2} + \frac{2B'}{r} e^{-2B} \right] / (D-2)$$

$$= - \frac{(D-2)}{L^2}$$

$$\text{or } \left[(D-3) \left(1 - e^{-2B} \right) \frac{1}{r} + 2B' e^{-2B} \right] = - \frac{(D-2)}{L^2} r$$

$$= \left[\left(1 - e^{-2B} \right) \frac{(D-3)}{r} + \frac{d}{dr} \left(1 - e^{-2B} \right) \right] = - \frac{(D-2)}{L^2} r$$

Int. factor $\mu = r^{D-3}$

Take $B(r_0) = \infty$?? , $F(r) \equiv 1 - e^{-2B(r)}$

$$F(r) = \left(\frac{r_0}{r} \right)^{D-3} - \frac{r^2}{L^2}$$

$$e^{-2B} = 1 + \left(\frac{r}{L}\right)^2 - \left(\frac{r_0}{r}\right)^{D-3} \equiv h(r)$$

$$ds^2 = -e^{-2B} dt^2 + e^{2B} dr^2 + r^2 d\Omega_{(D-2)}^2$$

$\therefore$ AdS-Schwarzschild metric

$$ds^2 = -h(r) dt^2 + \frac{1}{h(r)} dr^2 + r^2 d\Omega_{(D-2)}^2$$

$\leadsto$ Note, in $h(r) \sim \left(\frac{r}{L}\right)^2$ is the gravitational potential of a point mass in flat D-dim spacetime.

$r_0 \propto$ (Mass of BH)

• $h(r) = 0 \Rightarrow$ has a single zero

• Singular at $r=0$

$R_{\text{inner}} R_{\text{outer}} \xrightarrow{r \rightarrow 0} \infty$

So $r=0$ is a curvature singularity.

* Zero of $h(r)$ is a regular pt, of the event horizon.

* In the limit $L \rightarrow \infty$: $ds_{\text{AdS}_0}^2$ becomes asymptotically flat and describes a BH in M^D

* In the limit $r_0 \rightarrow 0$: $ds_{\text{AdS}}^2 = -(1 + \frac{r^2}{L^2}) dt^2 + \dots$

* AdS_0 is only maximally symmetric, i.e.

$g_{ab} = e^a{}_{\mu} e^b{}_{\nu} \eta^{\mu\nu}$ iff $r_0 = 0$ i.e. the mass of the BH vanishes.

The Reissner-Nordström metric:

$$s = \frac{1}{2} \left(\frac{1}{2} \left(\frac{1}{2} \right) \left(\frac{1}{2} \right) \right) = \frac{1}{2} \left(\frac{1}{2} \right) \left(\frac{1}{2} \right)$$

$$\begin{aligned}
 \delta S &= \int d^D x \left[\frac{1}{2} \sqrt{-g} - \delta g \left(\frac{1}{2\kappa^2} R - \frac{1}{4} F_{\mu\nu} F^{\mu\nu} \right) \right. \\
 &\quad \left. + \left(\frac{1}{2\kappa^2} \delta R - \frac{1}{4} \delta (F_{\mu\nu} F^{\mu\nu}) \right) \right] \\
 &= \int d^D x \left[\left(\frac{1}{2\kappa^2} \sqrt{-g} g_{\mu\nu} \delta g^{\mu\nu} \right) R + \frac{1}{2\kappa^2} \delta g^{\mu\nu} R_{\mu\nu} \sqrt{-g} \right] \\
 &\quad - \frac{1}{4} \int d^D x \delta (F_{\mu\nu} F^{\mu\nu}) \\
 &= \int d^D x \sqrt{-g} \left(\frac{1}{2\kappa^2} \right) \left[R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R \right] \delta g^{\mu\nu} \\
 &\quad - \frac{1}{4} \int d^D x \delta (F_{\mu\nu} F^{\mu\nu}) + \frac{1}{8} \int d^D x \sqrt{-g} g_{\mu\nu} \delta g^{\mu\nu} F_{\sigma\lambda} F^{\sigma\lambda}
 \end{aligned}$$

$$\begin{aligned}
 &\hookrightarrow = -\frac{1}{4} \int d^D x \delta g^{\mu\sigma} g^{\nu\lambda} F_{\mu\nu} F_{\sigma\lambda} \\
 &= -\frac{1}{4} \int d^D x \left[\delta g^{\mu\sigma} g^{\nu\lambda} F_{\mu\nu} F_{\sigma\lambda} + g^{\mu\sigma} \delta g^{\nu\lambda} F_{\mu\nu} F_{\sigma\lambda} \right] \\
 &= -\frac{1}{4} \int d^D x \left[\delta g^{\mu\sigma} F_{\mu\lambda} F_{\sigma}{}^{\lambda} + \delta g^{\nu\lambda} F_{\lambda\mu} F_{\sigma}{}^{\mu} \right] \\
 &= -\frac{1}{4} \int d^D x \left[\delta g^{\mu\nu} \left[F_{\mu\lambda} F_{\nu}{}^{\lambda} + F_{\lambda\mu} F_{\nu}{}^{\lambda} \right] \right] \\
 &= -\frac{1}{2} \int d^D x \left[F_{\mu\lambda} F_{\nu}{}^{\lambda} \right] \delta g^{\mu\nu}
 \end{aligned}$$

$$\begin{aligned}
 &= \int d^D x \sqrt{-g} \left[\left(\frac{1}{2\kappa^2} \right) \left(R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R \right) - \frac{1}{2} F_{\mu\lambda} F_{\nu}{}^{\lambda} + \frac{1}{8} g_{\mu\nu} F_{\sigma\lambda} F^{\sigma\lambda} \right] \delta g^{\mu\nu} \\
 &\therefore \boxed{\frac{1}{\kappa^2} \left(R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R \right) = F_{\mu\lambda} F_{\nu}{}^{\lambda} - \frac{1}{4} g_{\mu\nu} F_{\sigma\lambda} F^{\sigma\lambda}}
 \end{aligned}$$

where the energy momentum tensor:

$$\boxed{T_{\mu\nu} = F_{\mu\lambda} F_{\nu}{}^{\lambda} - \frac{1}{4} g_{\mu\nu} F_{\sigma\lambda} F^{\sigma\lambda}}$$



or,

$$R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R = T_{\mu\nu} \kappa^2 \Rightarrow R - \frac{D}{2} R = T$$

$$R = \left(T / \left(1 - \frac{D}{2} \right) \right) \kappa^2 = \frac{2T\kappa^2}{(2-D)}$$

$$R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} \left(\frac{2T\kappa^2}{2-D} \right) = \kappa^2 T_{\mu\nu}$$

$$\boxed{R_{\mu\nu} = \kappa^2 \left[T_{\mu\nu} - \frac{1}{(D-2)} g_{\mu\nu} T^{\rho}{}_{\rho} \right]}$$



$$\partial_{\mu} (\sqrt{-g} g^{\mu\rho} g^{\nu\sigma} F_{\rho\sigma}) = 0$$

$$R_{\mu\nu} = \kappa^2 \left[T_{\mu\nu} - \frac{1}{(D-2)} g_{\mu\nu} T^{\rho}{}_{\rho} \right]$$

$$T_{\mu\nu} = F_{\mu\rho} F_{\nu}{}^{\rho} - \frac{1}{4} g_{\mu\nu} F_{\rho\sigma} F^{\rho\sigma}$$

$$\text{Now } g^{\sigma\mu} R_{\mu\nu} = K^2 \left[T_N^\sigma - \frac{\delta_\nu^\sigma}{(D-2)} T_\rho^\rho \right]$$

$$R_N^\mu = K^2 \left[T_N^\mu - \frac{\delta_\nu^\mu}{(D-2)} T_\rho^\rho \right]$$

$$\bullet T = g^{\mu\nu} F_{\mu\rho} F_{\nu\rho} - \frac{D}{4} F^2 = F^2 \left(1 - \frac{D}{4} \right) \\ = (F^2/4)(4-D)$$

$$T_N^\mu = F^\mu_\rho F_N^\rho - \frac{1}{4} \delta_N^\mu F^2$$

$$R_N^\mu = K^2 \left[F^\mu_\rho F_N^\rho - \frac{1}{4} \delta_N^\mu F^2 + \frac{\delta_N^\mu (D-4)}{(D-2)4} F^2 \right] \\ = K^2 \left[F^\mu_\rho F_N^\rho + \left(\delta_N^\mu \frac{1}{4} F^2 \right) \left(\frac{D-4}{(D-2)} - 1 \right) \right]$$

$$R_N^\mu = K^2 \left[F^\mu_\rho F_N^\rho - \delta_N^\mu \frac{F^2}{2(D-2)} \right]$$

* The only non-vanishing components of F are $F_{tr} = -F_{rt} = f(r)$

$$\therefore F_{\rho\sigma} g^{\rho\lambda} g^{\sigma\mu} F_{\lambda\mu} = F_{t\sigma} g^{t\lambda} g^{\sigma\mu} F_{\lambda\mu} + F_{r\sigma} g^{\sigma\lambda} g^{\mu\lambda} F_{\lambda\mu} \\ = F_{tr} (-e^{-2A}) g^{\sigma\mu} F_{t\mu} + F_{r\sigma} e^{2B} g^{\sigma\mu} F_{r\mu} \\ = (F_{tr})^2 (-e^{-2A}) e^{-2B} + F_{rt} e^{2B} (-e^{-2A}) F_{rt}$$

$$\therefore F_{\rho\sigma} F^{\rho\sigma} = -2f^2(r) e^{-2(A+B)}$$

$$F_{\mu\rho} F_{\nu}^\rho = F_{\mu t} F_{\nu}^t + F_{\mu r} F_{\nu}^r = (-e^{-2A}) F_{\mu t} F_{\nu}^t + e^{-2B} F_{\mu r} F_{\nu}^r$$

$$(FF)_{tt} = e^{-2B} f^2 (-e^{-2A}) = -(FF)_t^t$$

$$(FF)_{rr} = -e^{-2A} f^2 (e^{-2B}) = -(F^r F_r)$$

$$R_t^t = K^2 \left(-e^{-2(A+B)} f^2 - \frac{(-2f^2 e^{-2(A+B)})}{2(D-2)} \right) \\ = K^2 e^{-2(A+B)} f^2 \left[\frac{1}{(D-2)} - 1 \right] \\ = -K^2 e^{-2(A+B)} f^2 \frac{(D-3)}{(D-2)}$$

$$R_r = R_t \quad \checkmark \quad \Rightarrow \quad A' = -B'$$

$$\partial_\mu \left((-r^{D-2}) g^{\mu\rho} g^{\nu\sigma} F_{\rho\sigma} \right) = 0$$

$$\partial_t \left((r^{D-2}) g^{tt} g^{r\sigma} F_{t\sigma} + g^{tr} g^{\nu\sigma} F_{r\sigma} + g^{tt} g^{\nu\sigma} F_{t\sigma} \right)$$

$$\partial_t (r^{D-2} g^{tt} g^{r\sigma} F_{t\sigma}) + \partial_r (e^{-2B} g^{\nu\sigma} F_{r\sigma}) = 0$$

$$\partial_r (e^{-2B} r^{D-2} g^{tt} F_{rt}) = 0$$

$$\partial_r (r^{D-2} e^{2(B+A)} F_{rt}) = 0$$

$$\text{or } \partial_r (r^{D-2} f(r)) = 0$$

$$r^{D-2} f(r) = \tilde{q} \Rightarrow f(r) = \frac{\tilde{q}}{r^{D-2}}$$

$$\text{Normalise: } \tilde{q} = -q / \Omega_{D-2}$$

$$F_{tr} = f(r) = -\frac{q}{\Omega_{D-2} r^{D-2}}$$

$$\text{for } D=4: \quad \Omega_2 = \frac{2\pi^{3/2}}{\Gamma(3/2)} = \frac{2\pi^{3/2}}{\frac{1}{2}\Gamma(1/2)} = 4\pi$$

$$R_j^i = \delta_j^i \left[(D-3) (1 - e^{2B}) \frac{1}{r^2} + \frac{2B'}{r} e^{-2B} \right]$$

and

$$R_j^i = \delta_j^i \left(-K^2 \frac{F^2}{2(D-2)} \right) = \delta_j^i \left(\frac{K^2}{2(D-2)} \right) (2f^2)$$

$$= \delta_j^i \frac{K^2 q^2}{(D-2)(\Omega_{D-2} r^{D-2})^2}$$

solving for $F(r) \equiv e^{-2B}$:

$$F(r) \equiv e^{-2B} = 1 - \left(\frac{r_0}{r} \right)^{D-3} + \frac{K^2 q^2}{(D-2)(D-3)(\Omega_{D-2} r^{D-2})^2}$$

Thus, we have the Reissner-Nordström metric in D-dim spacetime:

$$ds^2 = -F(r) dt^2 + \frac{1}{F(r)} dr^2 + r^2 d\Omega_{(D-2)}^2$$

$$D=4: \quad F(r) = 1 - \frac{2MG}{r} + \frac{q^2 G}{4\pi r^2}$$

→ Roots of the $F(r)$:

$$r = r_{\pm} = MG \pm \sqrt{M^2 G^2 - \frac{q^2 G}{4\pi}}$$

→ Singularity: $r = 0$

This is a physical singularity since $K \rightarrow \infty$ for $r \rightarrow 0$

(i) For small charges of q : $q^2 < 4\pi M^2 G$

$r_{\pm}$ are real: Inner & outer horizons

- $r_{\pm}$: Event Horizons

- $r_{\pm}$: Both null hypersurfaces

CD can reach r_{-} and the "fall" into a BH.

(ii) $q^2 > 4\pi M^2 G$: $r_{\pm} \in \mathbb{C}$ Naked Singularity
 - Unphysical by Cosmic Censorship Conjecture

(iii) $q^2 = 4\pi M^2 G$: $r_{+} = r_{-}$ [Extremal Reissner-Nordström]

$$ds^2 = -\left(1 - \frac{MG}{r}\right)^2 dt^2 + \left(1 - \frac{MG}{r}\right)^{-2} dr^2 + r^2 d\Omega_2^2$$

- DBPS solⁿ of supergravity.

Horizon: $r = MG$

* We can shift the radial coordinate so that $n = 0$ at the horizon:

$$n = r - MG$$

$$ds^2 = -\left(1 + \frac{MG}{n}\right)^{-2} dt^2 + \left(1 + \frac{MG}{n}\right)^2 [dn^2 + n^2 d\Omega_2^2]$$

- Near horizon limit $n \rightarrow 0$

$$(i) \frac{n^2}{(n+MG)^2} = \left(\frac{n}{MG}\right)^2 \frac{1}{\left(\frac{n}{MG} + 1\right)^2} = \left(\frac{n}{MG}\right)^2 (1 + \mathcal{O}(n))$$

$$(ii) \frac{(MG)^2}{r^2} \left(\frac{r^2 + a^2}{MG} \right)^2 \approx \frac{(MG)^2}{r^2}$$

↳ Thus we find:

$$ds^2 \approx - \frac{r^2}{(MG)^2} dt^2 + (MG)^2 \frac{dr^2}{r^2} + (MG)^2 d\Omega_2^2$$

Finally, switching to the coordinates

$$z := \frac{(MG)^2}{r}, \quad dz = - \frac{(MG)^2}{r^2} dr$$

$$dz = - \frac{z^2}{(MG)^2} dr \Rightarrow dr = - \left(\frac{1}{z^2} \right) (MG)^2 dz$$

$$(dr)^2 = \frac{1}{z^4} (MG)^4 dz^2$$

$$\frac{1}{r^2} = \frac{z^2}{(MG)^4}$$

$$\Rightarrow \left(\frac{dr}{r^2} \right) (MG)^2 = \left(\frac{z^2 (MG)^4 dz^2}{z^4 (MG)^4} \right) (MG)^2 = \frac{(MG)^2}{z^2} dz^2$$



$$\Rightarrow ds^2 \approx \frac{(MG)^2}{z^2} [-dt^2 + dz^2] + (MG)^2 d\Omega_2^2$$



$$z = (MG)^2/r, \quad r = r - MG$$

Recall that the Poincaré patch was

$$ds^2 = \frac{L^2}{z^2} [dz^2 - dt^2 + \sum_i (dx^i)^2]$$

• For AdS_2 :

$$ds^2 = \frac{L^2}{z^2} [dz^2 - dt^2]$$

* Thus this metric is the direct product

$$AdS_2 \times S^2$$

(i) AdS Length scale: $L = MG$

(ii) Radius of S^2 : $r_s = MG$

: Robinson - Bertotti metric

A more general Rossner - Nordström solution:

Define:

$$\begin{aligned}
 F_{\mu\nu}^+ &= qF_{\mu\nu} + ip\tilde{F}_{\mu\nu} = qF_{\mu\nu} + p\tilde{F}_{\mu\nu} \\
 F_{\mu\nu}^+ &= \frac{1}{2}(F_{\mu\nu} + i\tilde{F}_{\mu\nu}) = \frac{1}{2}(qF_{\mu\nu} + ip\tilde{F}_{\mu\nu} + i(q\tilde{F}_{\mu\nu} - pF_{\mu\nu})) \\
 &= \frac{1}{2}(qF_{\mu\nu} + ip\tilde{F}_{\mu\nu} + iq\tilde{F}_{\mu\nu} - pF_{\mu\nu}) \\
 &= (qF_{\mu\nu}^+ + ip\tilde{F}_{\mu\nu}^+) \\
 &= (qF_{\mu\nu}^+ + ipF_{\mu\nu}^+) \\
 \boxed{F_{\mu\nu}^+} &= (q + ip)F_{\mu\nu}^+
 \end{aligned}$$

Claim: The stress tensor $T_{\mu\nu}(F)$ is the same as F w/ a const. $(p^2 + q^2)$

$$T_{\mu\nu}(F) \equiv F_{\mu\rho}F_{\nu}^{\rho} - \frac{1}{4}g_{\mu\nu}F_{\rho\sigma}F^{\rho\sigma} = (p^2 + q^2)T_{\mu\nu}(F)$$

This means the gravitational effects of the two configurations are identical.

Proof:

$$\begin{aligned}
 & (qF_{\mu\rho} + p\tilde{F}_{\mu\rho})(qF_{\nu}^{\rho} + p\tilde{F}_{\nu}^{\rho}) \\
 &= q^2 F_{\mu\rho}F_{\nu}^{\rho} + pqi\tilde{F}_{\mu\rho}F_{\nu}^{\rho} + ipqF_{\mu\rho}\tilde{F}_{\nu}^{\rho} - p^2\tilde{F}_{\mu\rho}\tilde{F}_{\nu}^{\rho}
 \end{aligned}$$

$$\textcircled{a} F_{\mu\rho}\tilde{F}_{\nu}^{\rho} = \frac{1}{4}g_{\mu\nu}F_{\rho\sigma}\tilde{F}^{\rho\sigma}$$

$$\textcircled{b} \tilde{F}_{\mu\rho}\tilde{F}_{\nu}^{\rho} = -F_{\mu\rho}F_{\nu}^{\rho} + \frac{1}{2}g_{\mu\nu}F_{\rho\sigma}F^{\rho\sigma}$$

$$\begin{aligned}
 & -\frac{1}{4}g_{\mu\nu}F_{\rho\sigma}F^{\rho\sigma} = -\frac{1}{4}g_{\mu\nu}(qF_{\rho\sigma} + ip\tilde{F}_{\rho\sigma})(qF^{\rho\sigma} + ip\tilde{F}^{\rho\sigma}) \\
 &= \left(\frac{1}{4}g_{\mu\nu}\right)[q^2 F_{\rho\sigma}F^{\rho\sigma} + ipq\tilde{F}_{\rho\sigma}F^{\rho\sigma} + ipqF_{\rho\sigma}\tilde{F}^{\rho\sigma} - p^2\tilde{F}_{\rho\sigma}\tilde{F}^{\rho\sigma}]
 \end{aligned}$$

$$\begin{aligned}
 &= q^2 F_{\mu\rho}F_{\nu}^{\rho} + \frac{1}{2}pq\tilde{F}_{\mu\rho}F_{\nu}^{\rho} + p^2 F_{\mu\rho}F_{\nu}^{\rho} - \frac{1}{2}p^2 \underbrace{g_{\mu\nu}F_{\rho\sigma}F^{\rho\sigma}}_{\text{const.}}
 \end{aligned}$$

$$\begin{aligned}
 &= -\frac{1}{4}g_{\mu\nu}q^2 F^2 - \left(\frac{1}{2}g_{\mu\nu}\right)(ipq)\tilde{F}_{\rho\sigma}\tilde{F}^{\rho\sigma} + \frac{1}{4}g_{\mu\nu}p^2 F_{\rho\sigma}F^{\rho\sigma}
 \end{aligned}$$

$$\Sigma = (q^2 + p^2)F_{\mu\rho}F_{\nu}^{\rho} - \frac{1}{4}g_{\mu\nu}F_{\rho\sigma}F^{\rho\sigma} - \frac{1}{4}g_{\mu\nu}q^2 F_{\rho\sigma}F^{\rho\sigma}$$

$$T_{\mu\nu}(F) = (p^2 + q^2) T_{\mu\nu}(F)$$

* For a unit charge $q=1$, $F^{tr} = -F_{tr} = \frac{2}{4\pi r^2}$
 v/ its dual $\tilde{F}_{\theta\phi} = -i(\sin\theta/4\pi)$ | unit mag. charge

• For $r > 0$:

$$\partial_\mu (\sqrt{g} * F^{\mu\nu}) = 0$$

$$\partial_\mu * F_{\mu\nu} = 0$$

$$\Rightarrow \text{Metric: } F(r) = 1 - \frac{2mG}{r} + \frac{(q^2 + p^2)G}{4\pi r^2}$$

$$q^2 + p^2 \leq 4\pi M^2 G$$

Killing Spinors & BPS Sol^{ns}:

* Killing spinors are associated with a special class of sol^{ns} of the E.O.M. of Supergravity theory.

→ They are a finite subset of the spinor χ^a for which the SUSY transf. leaves the sol^{ns} invariant: i.e. unchanged from its original form.

- These spinors contain a finite number of constant parameters which determine the residual global Supersymmetry algebra of the solution

* Killing Spinor conditions come from fermion. transf. rules of sugra.

→ Killing spinor sol^{ns} are similar to BPS magnetic monopole sol^{ns}.

* For $(N=1, D=4)$:

(i) Sol^{ns} (classical) is M^4 $(g_{\mu\nu})_0 = \eta_{\mu\nu}$ with vanishing gravitino field $(\psi_\mu)_0 = 0$. I.e., it's torsion-free.

The Action $R = e^{a\mu} e^{b\nu} R_{\mu\nu ab}$

$$S_s = \frac{1}{2\kappa^2} \int d^4x (R - \bar{\psi}_\mu \gamma^{\mu\nu\rho} \partial_\nu \psi_\rho)$$

is invariant under the susy:

$$\delta e_\mu^a = \frac{1}{2} \bar{\epsilon} \gamma^a \psi_\mu$$

$$\delta \psi_\mu = D_\mu \epsilon(x) \equiv \partial_\mu \epsilon + \frac{1}{4} \omega_{\mu ab} \gamma^{ab} \epsilon$$

• In the most general of cases we have an extra torsion term in $D_\mu \epsilon \sim (\psi_\mu)^2$

$$\omega_{\mu ab} = \omega_{\mu ab}(e) + k_{\mu ab}$$

$$k_{\mu\nu\rho} = -\frac{1}{6} (\bar{\psi}_\mu \gamma_\rho \psi_\nu - \bar{\psi}_\nu \gamma_\mu \psi_\rho + \bar{\psi}_\rho \gamma_\nu \psi_\mu)$$

→ The residual Global susy is determined? by the conditions:

$$\delta e_\mu^a = \frac{1}{2} \bar{\epsilon} \gamma^a \psi_\mu = 0 \quad \& \quad \delta \psi_\mu = D_\mu \epsilon = 0$$

$$(e_\mu^a)_0 = \delta_\mu^a \quad \text{and} \quad (\psi_\mu)_0 = 0$$

• The first is trivially satisfied, the second:

$$\partial_\mu \epsilon + \frac{1}{4} \omega_{\mu ab}(e) \gamma^{ab} \epsilon = 0$$

In flat spacetime $(g_{\mu\nu})_0 = \eta_{\mu\nu}$ we have

$\omega_{\mu ab} = 0$, thus, the second condition is

simply:

* We conclude: Killing spinors in a M^4 background are the set of four-independent constant Majorana spinors ϵ_α

$\Rightarrow$ We infer that the symmetry alg. of fluctuations about M^D is the $D=4$ Poincaré SUSY alg.

* For any two Killing spinors ϵ and ϵ^* , the bilinear $(\bar{\epsilon} \gamma \epsilon)$ is a Killing Vector (constant) and is thus a Killing vec. for spacetime translations.

$\Rightarrow$ For any given theory:

$B(x)$: Bosonic fields

$F(x)$: Fermionic fields

Then, the local susy:

$$\delta_Q B(x) = \bar{\epsilon}(x) f_1(B(x)) F(x) + \mathcal{O}(F^3)$$

$$\delta_Q F(x) = f_2(B(x)) \epsilon(x) + \mathcal{O}(F^2)$$

- For each preserved susy $\exists$ a nontrivial const. $\epsilon(x)$ for which δB and δF vanish when the classical solⁿ is substituted into $\delta_Q F$ and $\delta_Q B$

- Since, classically, the fermionic fields vanish $\delta B = 0$ trivially satisfied.

$\Rightarrow$ Only need to study the linearized ferm. variations:

$$\delta F(x)|_{cl} \equiv f_2(B(x)) \epsilon(x) = 0$$

- These are a set of coupled first order ODEs
- When a solution exists, it determines a set of n_Q linearly independent spinor configurations for $E(x)$
 - Each independent $E(x)$ is a Killing Spinor.
 - We say that the BPS solⁿ preserves n_Q supersymmetries

The integrability condition for Killing Spinors:

Killing Spinor condition

$$D_\mu E = 0$$

Noncommutativity:

$$[D_\mu, D_\nu] E = \frac{1}{4} R_{\mu\nu ab} \gamma^{ab} E = 0 \quad \left(\begin{array}{l} \text{Integrability} \\ \text{Condition} \end{array} \right)$$

Commuting and Anti-commuting Killing Spinors:

- Spinors in sugra are anti-commuting, so too are $E(x)$.

Killing Spinors are determined by:

$$\underline{\delta \psi_\mu = 0} \quad \text{and} \quad \underline{\delta \chi = 0}$$

evaluated in a purely bosonic field conf.

- This leads to linear eqs in E . No way of determining whether E are commuting or anti-commuting

* Thus, we can view the eq. as a set of ordinary differential in alg. conditions whose solution, when it exists, is expressed in terms of ordinary commuting quantities.

When the Killing spinors are applied to 12 vectors about BPS backgrounds, they must be converted to anti-commuting variables.

Killing Spinors and AdS:

- AdS: Solⁿ to sugra w/ $-\Lambda$, $N=1$. It also occurs for $D \geq 4$ in KK-reduction of 10 & 12-dim sugra

→ Find Killing spinors for AdS_D:

$$S = \frac{1}{2\kappa^2} \int d^D x (R - \Lambda)$$

In AdS_D, the susy transf. are

$$\delta e_m^a = \frac{1}{2} \bar{\epsilon} \gamma^a \psi_m \quad \text{and} \quad \delta \psi_m = \hat{D}_m \epsilon(x)$$

- Thus, the Killing condition is given by

$$\hat{D}_m \epsilon \equiv \left(D_m - \frac{1}{2L} \gamma_m \right) \epsilon = 0$$

the integrability condition:

$$\begin{aligned} [\hat{D}_m, \hat{D}_n] \epsilon &= \frac{1}{4} \left(R_{mnab}(\omega) + \frac{2}{L^2} (e_{ma} e_{nb} - e_{mb} e_{na}) \right) \gamma^{ab} \epsilon \\ &= \left(\frac{1}{4} R_{mnab} \gamma^{ab} + \frac{2}{2L^2} \gamma_{mn} \right) \epsilon \end{aligned}$$

Since AdS_D is maximally symmetric

$$R_{mnpq} = \frac{R}{D(D-2)} (g_{mp} g_{nq} - g_{mq} g_{np})$$

But

$$R_{mn} = -\left(\frac{D-2}{2^2}\right) g_{mn} \Leftrightarrow \underline{R = -\frac{(D-2)D}{L^2}}$$

$$\therefore R_{mnpq} = -\frac{1}{L^2} (g_{mp} g_{nq} - g_{mq} g_{np})$$

$$R_{mnab} = -\frac{1}{L^2} (e_{ma} e_{nb} - e_{mb} e_{na})$$

$$\therefore \underline{[\hat{D}_m, \hat{D}_n] \epsilon = 0}$$

And so AdS_D is maximally supersymmetric w/ a full $2^{\lfloor D/2 \rfloor}$ Killing Spinors

Claim: Assume ϵ and ϵ' are both Killing Spinors.

Then $\nabla_\mu (\bar{\epsilon}' \gamma_\mu \epsilon) = -\left(\frac{2}{L}\right) \bar{\epsilon}' \gamma_\mu \epsilon$

and the vector $K_\mu \equiv \bar{\epsilon}' \gamma_\mu \epsilon$ satisfies

$$\nabla_\mu K_\nu + \nabla_\nu K_\mu = 0$$

Proof: $\nabla_\mu \bar{\epsilon}' \gamma_\mu \epsilon + \bar{\epsilon}' \gamma_\mu \nabla_\mu \epsilon$ (tot. = -2)

$$= \nabla_\mu \bar{\epsilon}' \gamma_\mu \epsilon + \bar{\epsilon}' \gamma_\mu \nabla_\mu \epsilon$$

$$= \nabla_\mu \bar{\epsilon}' \gamma_\mu \epsilon + \bar{\epsilon}' \gamma_\mu \left(\frac{1}{2L} \gamma_\mu\right) \epsilon$$

where

$$\begin{aligned} \nabla_\mu \bar{\epsilon}' &= \overline{\frac{1}{2L} (\gamma_\mu \epsilon')} = \frac{1}{2L} (\gamma_\mu \epsilon')^T C \\ &= \frac{1}{2L} (\epsilon')^T \gamma_\mu^T C = \left(\frac{1}{2L}\right) (\epsilon')^T (t_0 t_1) C \gamma_\mu C^{-1} C \\ &= -\frac{1}{2L} \bar{\epsilon}' \gamma_\mu \end{aligned}$$

$$\begin{aligned} \therefore &= -\frac{1}{2L} \bar{\epsilon}' \gamma_\mu \gamma_\nu \epsilon + \bar{\epsilon}' \gamma_\mu \gamma_\mu \left(\frac{1}{2L}\right) \epsilon \\ &= -\frac{1}{2L} (\bar{\epsilon}' (\gamma_\mu \gamma_\nu - \gamma_\nu \gamma_\mu) \epsilon) \end{aligned}$$

$$\therefore \nabla_\mu (\bar{\epsilon}' \gamma_\mu \epsilon) = -\frac{2}{L} \bar{\epsilon}' \gamma_\mu \epsilon$$

Now, define $K_\mu \equiv \bar{\epsilon}' \gamma_\mu \epsilon$

$$\nabla_\mu K_\nu = -\frac{1}{L} \bar{\epsilon}' \gamma_{\mu\nu} \epsilon$$

$$\nabla_\nu K_\mu = -\frac{1}{L} \bar{\epsilon}' \gamma_{\nu\mu} \epsilon = \frac{1}{L} \bar{\epsilon}' \gamma_{\mu\nu} \epsilon$$

$$\therefore \nabla_\mu K_\nu + \nabla_\nu K_\mu = 0 \quad \square$$

* We will study Killing Spinors in the Poincare Patch of AdS_D : $ds^2 = \frac{L^2}{z^2} (dz^2 - dt^2 + \sum_{i=1}^{D-2} (dx^i)^2)$

$\rightarrow$ Change the radial coordinate $z = L e^{-r/L}$

such that the Boundary appears in the limit $r \rightarrow \infty$.

$$dz = -e^{-r/L} dr \Rightarrow (dz)^2 = e^{-2r/L} dr^2$$

$$ds^2_{AdS_D} = e^{2r/L} \left[e^{-2r/L} dr^2 - dt^2 + \sum_{i=1}^{D-2} (dx^i)^2 \right]$$

$$x^m: m=0, \dots, D-2$$

• Choose 1-frames

$$e^{\hat{a}} \equiv e^{r/L} dx^m$$

$$e^r \equiv dr$$

$$\omega^{\hat{a}r} = \frac{1}{L} e^{\hat{a}}$$

$$\omega^{\hat{a}\hat{b}} = 0$$

Claim: The 2-form ρ^{ab} is maximally symmetric
i.e. $\rho^{ab} = -e^a \wedge e^b / L^2$

Proof: $\rho^{ab} = d\omega^{ab} + \omega^a{}_c \wedge \omega^{cb}$
 $d\omega^{ab} = d(\omega^{\hat{a}r}) = \frac{1}{L} d(e^{r/L} dx^m) = \frac{e^{r/L}}{L^2} dr \wedge dx^m$
 $= (\frac{1}{L^2}) e^r \wedge e^{\hat{a}}$
 $\omega^a{}_c \wedge \omega^{cb} = \omega^{\hat{a}}{}_{\hat{r}} \wedge \omega^{\hat{r}b} + \omega^{\hat{a}}{}_r \wedge \omega^{rb}$
 $\omega^{\hat{a}}{}_{\hat{r}} \wedge \omega^{\hat{r}b} + \omega^{\hat{a}}{}_r \wedge \omega^{rb}$

Thus $\rho^{ab} = -\frac{1}{L^2} (e^a \wedge e^b)$

• f^a : Flat D-dim spacetime $| \gamma^r = r$
 $\gamma^{\hat{a}}, \hat{a}=0, 1, \dots, D-2$

→ The Killing Spinor Conditions split into radial and transverse components:

$$\hat{D}_r E = (\partial_r - \frac{1}{2L} \gamma_r) E = 0$$

$$\hat{D}_\mu E = (\partial_\mu + \frac{1}{2L} \gamma_\mu (\gamma_r - 1)) E = 0$$

Proof: $\hat{D}_r E = (\partial_r - \frac{1}{2L} \gamma_r) E$ (i)

But $\omega_r^{ab} = 0$, thus

$$\hat{D}_r E = (\partial_r - \frac{1}{2L} \gamma_r) E$$

(ii) $\hat{D}_\mu E = (\partial_\mu - \frac{1}{2L} \gamma_\mu) E$

$$= (\partial_\mu + \frac{1}{4} \omega_\mu^{ab} \gamma_{ab} - \frac{1}{2L} \gamma_\mu$$

$$\omega_\mu^{ab} \gamma_{ab} = \omega_\mu^{\hat{a}b} \gamma_{\hat{a}b} + \omega_\mu^{r\hat{b}} \gamma_{r\hat{b}}$$

$$= \omega_\mu^{\hat{a}r} \gamma_{\hat{a}r} + \omega_\mu^{r\hat{b}} \gamma_{r\hat{b}}$$

$$= 2\omega_{\mu}^{\hat{m}r} \gamma_{\hat{m}r}$$

$$\Rightarrow \hat{D}_{\mu} \epsilon = \left(\partial_{\mu} + \frac{1}{4} (2\omega_{\mu}^{\hat{m}r} \gamma_{\hat{m}r}) - \frac{1}{2L} \gamma_{\mu} \right) \epsilon$$

$$= \left(\partial_{\mu} + \frac{1}{2} \omega_{\mu}^{\hat{m}r} \gamma_{\hat{m}r} - \frac{1}{2L} \gamma_{\mu} \right) \epsilon$$

$$= \left(\partial_{\mu} + \frac{1}{2L} \gamma_{\mu} (\gamma_r - 1) \right) \epsilon$$

* Introduce constant spinors $\underline{n}_{\pm}$ s.t.

$$\gamma_r \underline{n}_{\pm} = \underline{n}_{\pm}$$

Then, the Killing Spinors are:

$$\epsilon_+ = e^{r/2L} \underline{n}_+$$

$$\epsilon_- = \left(e^{-r/2L} + \frac{1}{L} e^{r/2L} x^{\mu} \gamma_{\mu} \right) \underline{n}_-$$

$\Rightarrow$ first is trivial,

$$\partial_r \epsilon_- = \left(-\frac{1}{2L} e^{r/2L} + \frac{1}{2L^2} e^{r/2L} x^{\mu} \gamma_{\mu} \right) \underline{n}_-$$

$$\frac{1}{2L} \gamma_r \epsilon_- = -\frac{1}{2L} \epsilon_-$$

$$\Rightarrow -\frac{1}{2L} \left(e^{r/2L} + \frac{1}{L} e^{r/2L} x^{\mu} \gamma_{\mu} \right) \underline{n}_-$$

$$\partial_{\mu} \epsilon_- = \frac{1}{L} e^{r/2L} \gamma_{\mu} \underline{n}_-$$

$$-\frac{1}{2L} \gamma_{\mu} (\gamma_r - 1) \epsilon_- = \left(\frac{1}{L} \right) \gamma_{\mu} e^{r/2L}$$

Aside: AdS_5 isometries & Killing Spinors

$$\eta^M = \bar{\epsilon} \gamma^M \epsilon, \quad M := (m, r)$$

$$\delta Z = -Z \frac{\delta r}{L}$$

Define: $\delta X^M \equiv \bar{\epsilon} \gamma^M \epsilon$

(i) Translations: $x^{\mu} \rightarrow x^{\mu} + a^{\mu} \mid \delta r = 0$

$$\delta X^{\mu} = a^{\mu} \equiv \bar{\epsilon} \gamma^{\mu} \epsilon$$

Since $\bar{n}_{\pm} \gamma_r = \mp \bar{n}_{\pm}$ we find that

$$\bar{n}_{+} n_{+} = 0$$

$$\delta X^{\mu} = \left(e^{r/2L} n_{+} \right) \gamma^{\mu} \left(e^{r/2L} n_{+} \right) = e^{r/L} n_{+} \gamma^{\mu} n_{+}$$

Q: Not sure what happens to $e^{r/L}$ term

LA: It gets absorbed into $e^{\mu} \equiv e^{\mu} dx^{\mu}$

(ii) Lorentz:

$$\delta x^{\sigma} = \lambda^{\alpha\beta} (m_{\alpha\beta})^{\sigma}_{\lambda} x^{\lambda}$$

$$\bar{E}' \gamma^{\sigma} E = \bar{E}'_{\pm} \gamma^{\sigma} E_{\pm}$$

Note: E_{-} has a factor of $x^{\mu} \gamma_{\mu}$

and so it must have one E_{-} and one E_{+} to $\sim \lambda^{\mu} x^{\mu}$

$$\therefore \delta x^{\mu} = \bar{E}_{-} \gamma^{\mu} E_{+} + \bar{E}_{+} \gamma^{\mu} E_{-}$$

$$E_{-} \gamma^{\mu} E_{+} = (\bar{e}^{r/2L} \bar{n}_{-} + \frac{1}{2} x^{\alpha} \bar{n}_{-} \gamma_{\alpha}) \gamma^{\mu} e^{r/2L} n_{+}$$

$$\bar{E}_{+} \gamma^{\mu} E_{-} = e^{r/2L} \bar{n}_{+} \gamma^{\mu} (\bar{e}^{r/2L} + \frac{1}{2} e^{r/2L} x^{\sigma} \gamma_{\sigma}) n_{-}$$

$$= \Sigma = \bar{e}^{r/2L} \bar{n}_{-} \gamma^{\mu} e^{r/2L} n_{+} + \bar{e}^{r/2L} \bar{n}_{+} \gamma^{\mu} e^{r/2L} n_{-}$$

+ ...

~~$$\text{By } \bar{n}_{-} \gamma^{\mu} n_{+} = -\bar{n}_{+} \gamma^{\mu} n_{-} \text{ so } \bar{e}^{r/2L} \bar{n}_{-} \gamma^{\mu} e^{r/2L} n_{+} = -\bar{e}^{r/2L} \bar{n}_{+} \gamma^{\mu} e^{r/2L} n_{-}$$~~

$$\frac{1}{2} (x^{\sigma} \bar{n}_{-} \gamma_{\sigma} \gamma^{\mu} n_{+} + \bar{n}_{+} \gamma^{\mu} x^{\sigma} \gamma_{\sigma} n_{-})$$

$$= \frac{1}{2} (\bar{n}_{-} \gamma^{\sigma} \gamma^{\mu} n_{+} + \bar{n}_{+} \gamma^{\mu} \gamma^{\sigma} n_{-}) x_{\sigma}$$

$$= \frac{1}{2} (\bar{n}_{-} (\gamma^{\mu} \gamma^{\sigma} + \gamma^{\sigma} \gamma^{\mu} - \gamma^{\mu} \gamma^{\sigma}) n_{+} + \bar{n}_{+} (\gamma^{\mu} \gamma^{\sigma} - \gamma^{\sigma} \gamma^{\mu}) n_{-})$$

$$= \frac{1}{2} 2 (\bar{n}_{-} \gamma^{\mu} \gamma^{\sigma} n_{+} + \bar{n}_{-} \gamma^{\sigma} \gamma^{\mu} n_{+} + 2 \bar{n}_{+} \gamma^{\mu} \gamma^{\sigma} n_{-} + \bar{n}_{+} \gamma^{\sigma} \gamma^{\mu} n_{-})$$

$$= \frac{2}{2} (\bar{n}_{-} \gamma^{\sigma} \gamma^{\mu} n_{+} + \bar{n}_{+} \gamma^{\sigma} \gamma^{\mu} n_{-}) + \bar{n}_{-} \gamma^{\mu} \gamma^{\sigma} n_{+}$$

... Not too sure about this but this is the general idea of things (I think)
 $\rightarrow$ Idea: $\gamma^{\mu} \gamma^{\sigma} = \gamma^{\mu\sigma} + m^{\mu\sigma}$ so $m^{\mu\sigma}$ "variables" and is used for Scale trans.

(i) Translations: $\delta x^{\mu} = a^{\mu} = \bar{n}_{+} \gamma^{\mu} n_{+}$, $\delta z = 0$

(ii) Lorentz: $\delta x^{\mu} = \lambda^{\alpha\beta} x^{\alpha} \gamma^{\mu}_{\beta}$, $\lambda^{\alpha\beta} \equiv \frac{1}{2} (\bar{n}_{+} \gamma^{\alpha} \gamma^{\beta} n_{+} + \bar{n}_{-} \gamma^{\beta} \gamma^{\alpha} n_{-})$

$$\delta z = 0$$

(iii) Scale: $\delta x^{\mu} = \rho x^{\mu}$, $\delta z = \rho z$, $\rho = \frac{1}{2} (\bar{n}_{+} n_{-} - \bar{n}_{-} n_{+})$

$$\delta z = -2c \cdot x z \quad w/ \quad \underline{c^{\hat{n}}} = \frac{2}{L^2} \bar{n}' \cdot \hat{x}^{\hat{n}} n.$$

$\underline{n}_+$: Q-susy parameter

$\underline{n}_-$: S-susy parameter

Extremal Reissner-Nordström spacetime as BPS solutions:

* Look at BPS solutions to the Maxwell-Einstein system $S = \int d^4x \sqrt{g} \left(\frac{1}{2\kappa^2} R - \frac{1}{4} F_{\mu\nu} F^{\mu\nu} \right)$

Assume: $D=4$, t, x^i , $i=1,2,3$ and assume a static spacetime metric

$$\star ds^2 = -e^{2U(x)} dt^2 + e^{-2U(x)} dx^i dx^i$$

• Also assume non-vanishing electric field components: $F_{ti}(x) = -\partial_i A_t(x)$

$U(x)$: indicates this f^{μ} is independent of time, t .

* We have the Majumdar-Papapetrou solutions

$$e^{-U} = 1 + \sum_{I=2}^{\infty} \frac{M_I G}{|\vec{x} - \vec{x}_I|}$$

$$A_t = \pm \frac{q}{\sqrt{4\pi G}} e^U$$

Maxwells eqs of $\star$ are

$$\partial_\mu (\sqrt{-g} g^{\mu\rho} g^{\nu\sigma} F_{\rho\sigma}) = 0$$

$$\begin{aligned}
&= \partial_t (-g^{tt} g^{tt} F_{tt}) + \partial_i (F_{ij} g^{ij} g^{tt} F_{it}) \\
&= \partial_t (0) + \partial_i (e^{-2U} e^{2U} (-e^{2U}) F_{it}) \\
&= \partial_i (e^{-2U} \partial_i e^U) = -2 \partial_i U \partial_i U e^U + e^{2U} \partial_i (\partial_i U e^U) \\
&= -2 (\partial_i U)^2 e^U
\end{aligned}$$

...

$$\Rightarrow -\nabla^2 e^U = 0$$

$\therefore e^U$ is a harmonic function (wrt Cartesian coord) x^i .

Recall $\Delta := dd^\dagger + d^\dagger d : \Omega^r(M) \rightarrow \Omega^r(M)$

Since $d^\dagger : \Omega^{r+1} \rightarrow \Omega^r$

• If we choose:

$$e^U = 1 + \frac{MG}{|\vec{x}|}$$

then the metric becomes:

$$ds^2 = - \left(1 + \frac{MG}{|\vec{x}|}\right)^2 dt^2 + \left(1 + \frac{MG}{|\vec{x}|}\right)^2 dx^i dx^i$$

switching to $dx^i dx^i = dr^2 + r^2 d\Omega_2^2$

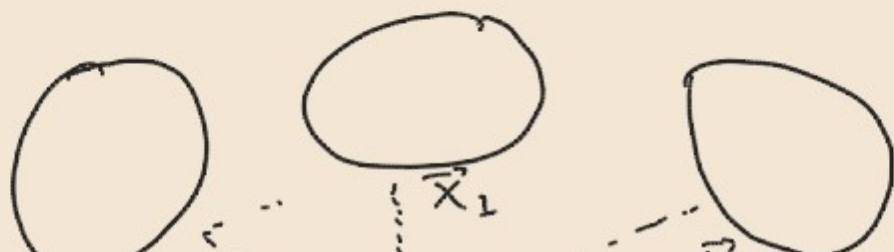
i.e. S^3 metric we obtain the extremal

Reissner-Nordström metric w/ $V \equiv |\vec{x}|$

and charge $q = \pm \sqrt{4\pi G M}$

* Note this is a ^{single} monopole solution.

More generally, a multi-center collection is given by $1 + \sum_{I=1}^n \frac{M_I G}{|\vec{x} - \vec{x}_I|}$ of n extremal Reissner-Nordström blackholes w/ Horizons located at $\vec{x} = \vec{x}_I$



Q: How do we define $\vec{x}_I$ as being a "horizon" is the horizon is "spread" along space (time) and is not singular.

→ System is stable since gravitational is electric repulsion cancel between pairs of BHs.

Idea: See how Papapetrou-Majumdar emerges as a BPS solⁿ of $N=2$ SUGRA.

SUGRA: $\Phi_{MN}, A_M, \underbrace{\psi_A^a}_{\text{two}}$

transformation rules contain two Maj. spinor parameters

* We work w/ the Chiral Projections of these Maj. spinors. So we use the UP/down position of the R-symmetry indices denoted: $A, B = 1, 2$

ex. For SUSY transf. parameters:

$$\begin{aligned} \gamma_* \epsilon^A &= \gamma_2 \epsilon^A = \epsilon^A & (L)_A \\ \gamma_* \epsilon_A &= -\gamma_R \epsilon_A = -\epsilon_A & (R)_A \end{aligned}$$

• Each spinor contains two complex components (or 4-real)

⇒ here, have 8-real parameters for $N=2$

Susy.

Killing

* Spinors are determined by the condition that the linearized gravitino trans. rule vanishes

$$\delta\psi_A = (D_m + \frac{1}{4}\gamma^{ab}W_{mab})\epsilon_A - \frac{1}{2}\sqrt{2}K\gamma^{ab}F_{ab}\epsilon_A D_m\epsilon^B$$

$$\therefore (D_m + \frac{1}{4}\gamma^{ab}W_{mab})\epsilon_A - \frac{1}{2}\sqrt{2}KF_{ab}\gamma^{ab}\epsilon_m\epsilon_A\epsilon^B = 0$$

• We can replace F_{ab} by F_{ab}^+ since $\delta\psi_A$ is chiral w/ P_R

and so

$$\gamma^{mn}F_{mn}P_L = \gamma^{mn}F_{mn}^-$$

$$\Rightarrow (\gamma^{mn}F_{mn}P_L)^{\dagger} = (\gamma^{mn})^{\dagger}F_{mn}^+$$

$$P_L^{\dagger}F_{mn}(\gamma^{mn})^{\dagger} = (\gamma^{mn})^{\dagger}F_{mn}^+$$

$$+ P_R F_{mn} \gamma^{mn} = \gamma^{mn} F_{mn}^+$$

* To analyze the condition:

Recall: $e^t = e^A dt$
 $e^r = e^B dr$

$$e^i = r \bar{e}^i$$

and: $\omega^{tr} = A' e^{-B} e^t$
 $\omega^{ij} = \bar{\omega}^{ij}$

$$\omega^{ir} = (e^{-B}/r) e^i$$

thus: $A = U$

$$\therefore \hat{e}^{\hat{0}} = e^U dt$$

$$\hat{e}^{\hat{i}} = e^{-U} dx^i$$

$$\Rightarrow \omega^{\hat{0}\hat{i}} = e^U \partial_i U \hat{e}^{\hat{0}}, \quad \omega^{\hat{i}\hat{j}} = -dx^i \partial_j U + dx^j \partial_i U$$

* We also note $\hat{F}_{\hat{0}\hat{i}} = F_{ti} = -\partial_i A_t$

$\leadsto$ The info we require is from the time

$$\delta\psi_{tA} = (\partial_t + \frac{1}{4}\gamma^{ab}\omega_{tab})\epsilon_A - \frac{1}{8\sqrt{2}K}F_{ab}\gamma^{ab}\gamma_t\epsilon_{AB}\epsilon^B = 0$$

$$\bullet \omega_{tab}\gamma^{ab} = 2e^{2U}\partial_i U \delta_{\hat{t}\hat{i}} \gamma^{\hat{0}\hat{i}} \epsilon_A = 2e^{2U}\partial_i U \left(\frac{1}{2}\right)(\gamma^0\gamma^i - \gamma^i\gamma^0)$$

$$\bullet F_{ab}\gamma^{ab} = 2(-\partial_i A_t)\gamma^0\gamma^i = \partial_i A_t 2$$

$$\therefore \delta\psi_{tA} = \partial_t \epsilon_A + \frac{1}{2}e^{2U}\partial_i U \gamma^{\hat{i}}\gamma^{\hat{0}}\epsilon_A - \frac{1}{4}\sqrt{2}K\partial_i A_t \gamma^{\hat{i}}\epsilon_{AB}\epsilon^B = 0$$

* Since sol^{ns} of interest are static, we assume that $\delta\psi_{tA}$ is independent of time. Thus $\epsilon_A \neq \epsilon_A(t)$ and so $\partial_t \epsilon_A = 0$

$\Rightarrow$ We can then see

$$e^{2U}\partial_i U \gamma^{\hat{i}}\gamma^{\hat{0}}\epsilon_A - \frac{1}{2}\sqrt{2}Ke^U\partial_i A_t \gamma^{\hat{i}}\epsilon_{AB}\epsilon^B = 0$$

$$e^U\partial_i U \gamma^{\hat{i}}\epsilon_A - \frac{1}{2}\sqrt{2}K\partial_i A_t \epsilon_{AB}\epsilon^B = 0$$

$$(K^{\frac{1}{2}}\sqrt{2}e^U\partial_i U)(\gamma^{\hat{i}}\epsilon_A) = (\partial_i A_t)(\epsilon_{AB}\epsilon^B)$$

$$\partial_i (K^{\frac{1}{2}}\sqrt{2}e^U)(\gamma^{\hat{i}}\epsilon_A) = \partial_i (A_t)(\epsilon_{AB}\epsilon^B)$$

$$\therefore A_t = \pm \sqrt{2}K^{-\frac{1}{2}}e^U \quad (\star)$$

$$\text{and } \gamma^{\hat{0}}\epsilon_A = \pm \epsilon_{AB}\epsilon^B \Leftrightarrow \epsilon_A = \mp \gamma^{\hat{0}}\epsilon_{AB}\epsilon^B$$

$$\epsilon_A = \mp \gamma^{\hat{0}}\epsilon_{AB}\epsilon^B \quad (\star)$$

